

Light Curve Asymmetry in Rubin LSST Variable Stars: Candidate Evidence for T^{0i} Momentum Flux Coupling in the Low-Velocity Galactic Regime

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Abstract

We report preliminary evidence from the live Rubin Observatory LSST alert stream (Days 6 to 8 of survey operations), cross-matched with Gaia DR3 proper motion vectors, for a directional photometric asymmetry in variable star light curves correlated with stellar proper motion. Across $n = 15$ matched Variable Star candidates, the primary finding is a speed-asymmetry correlation of $r = 0.726$, $p = 0.0022$, 3.06σ : faster-moving stars show systematically higher light curve asymmetry, with the highest proper-motion star ($PM = 30.0 \text{ mas/yr}$) showing the highest asymmetry (0.960). The relationship is monotonic, linear, and sky-wide. A second independent test, fall-rate suppression in high proper-motion stars ($p = 0.0433$, 2.02σ), supports the morphological prediction: fast-moving stars show fast rise and suppressed fall, consistent with a trailing wake geometry. Combined Fisher significance across both tests is 3.30σ , $p = 0.000977$. Supplementary Appendix A provides the formal mathematical proof that FFT-based N-body solvers structurally delete $\nabla \cdot T^{0i}$, with $\mathbb{E}[\lambda] \rightarrow 0$ confirmed empirically across 35 years of simulations (Figure 4). As a secondary and unexpected finding, an object with a notable positive residual from the speed-asymmetry trend sits at 21.5° from the CMB kinematic dipole ($l = 264^\circ$, $b = +48^\circ$), the top 1.1% of sky by proximity to that direction. An attempted ZTF confirmation failed on instrument-precision grounds, consistent with Rubin LSST being the uniquely capable instrument for this measurement. The theoretical interpretation and implications for galactic dynamics are discussed in Section 7.

The theoretical interpretation has been extended in four directions. First, the mechanism by which TreePM and FFT-based N-body solvers filter $\nabla \cdot T^{0i}$ below the force softening scale ε is made explicit, and the halo spin parameter $\lambda(\rho_{crit})$, falling monotonically from Barnes & Efstathiou (1987) through Benson (2017) to Li et al. (2022) across four orders of magnitude in density threshold, is identified as the accumulated numerical signature of this deletion. Second, the per-star coupling constant $g = d\alpha/dv$ measured here and the orbit-integrated coupling constant G measured in the companion SPARC paper are numerically consistent to within 16% ($g \cdot v / G = 1.16$ at $v = 30 \text{ km/s}$), connecting stellar photometry at pc scales to galactic rotation curves at kpc scales through the same T^{0i} proportionality. Third, this consistency generates a cross-paper falsifiable prediction: the α vs. μ relation should transition from linear to sublinear at a threshold set by the coherence ratio between stellar and galactic scales, predicted at $\mu_{crit} \approx 27 \text{ mas/yr}$ under equal coherence with no free parameters. Fourth, the radio continuum dipole anomaly, now exceeding 5σ across independent surveys (Böhme et al. 2025, PRL), is identified as a structural consequence of the same T^{0i} erasure: radio source counts couple to $T^{00} + T^{0i}$ while CMB temperature maps couple to T^{00} only, making the radio excess the expected signature of the missing momentum flux term rather than a survey systematic. The radio dipole anomaly and the missing mass problem may share the same root.

KEY RESULTS — n=15, Days 6–8 Rubin LSST

Measurement	Result
Speed-asymmetry correlation (primary)	$r = 0.726, p = 0.0022, 3.06\sigma$
Fall rate suppression (high-PM stars, $PM \geq 10$ mas/yr)	$t = -2.238, p = 0.0433, 2.02\sigma$
Combined significance (Fisher, 2 tests)	$3.30\sigma, p = 0.000977$
Fastest star → highest asymmetry	$PM = 30.0 \text{ mas/yr} \rightarrow \text{asym} = 0.960$
CMB-proximate object (Bubble 1)	$sep = 21.5^\circ$ — top 1.1% of sky (observation only)

Scientific Status: 3.30σ , $p = 0.000977$ combined candidate evidence across two independent tests. Not yet a confirmed detection. Speed-asymmetry and fall-rate suppression are both individually statistically significant. CMB proximity is noted as a secondary observation only not a formal statistical test.

1. Theoretical Motivation

The Vera C. Rubin Observatory produces an alert stream that includes Bazin light curve parameters for every detected transient — specifically `BBBRiseRate` and `BBBFallRate`, encoding the rise and fall timescales of each light curve. This paper asks a simple question: do faster-moving stars show more asymmetric light curves? If so, what is the physical origin of that asymmetry?

The theoretical framework motivating this question begins with Einstein’s field equations and the stress-energy tensor $T^{\mu\nu}$. This tensor has off-diagonal components T^{0i} that encode momentum flux i.e. the directed flow of energy-momentum through space due to bulk motion. For any gravitating body moving through a potential field, these terms are non-zero by construction. In standard cosmological N-body treatments, $T^{\mu\nu}$ is approximated as mass density ρ acting on the diagonal only, with all off-diagonal terms set to zero before any calculation begins. The observational prediction that follows from taking these terms seriously is direct: a star moving through the galactic potential generates a T^{0i} momentum flux in the direction of motion, creating a trailing wake that distorts the light curve asymmetrically. Rise rate exceeds fall rate. The degree of asymmetry scales with velocity. The Bazin parameterisation in the Rubin alert stream measures exactly this asymmetry.

The implications of a non-zero coupling for galactic dynamics including the possible relationship to the missing mass problem are deferred to Section 7, where the observational results can be evaluated on their own merits first. The present section establishes only that the theoretical framework makes a specific, testable, falsifiable prediction: asymmetry $\propto$ proper motion, with a morphology of fast rise and suppressed fall. That prediction is tested directly against data in Section 5.

2. Why Now

The Vera C. Rubin Observatory began full survey operations in early 2026[1]. Its 6.5-meter primary mirror, combined with a 3.2 gigapixel camera and five millimagnitude photometric precision, produces an alert stream of transient detections in near real time. Each alert includes Bazin light curve fit parameters[2], a parameterization originally developed for supernova rise and fall characterisation, including `BBBRiseRate` and `BBBFallRate`. Although the Bazin model was originally developed for supernovae, Rubin LSST applies the same rise and fall decomposition to all alerting sources, including variable stars. In this analysis, the Bazin rise and fall rates serve purely as pipeline provided shape descriptors, independent of any supernova specific physics. These two numbers, computed at detection time and included in every alert, are exactly the measurement the gravitational wake hypothesis requires.

No previous instrument provided this combination. ZTF, with its 1.2-meter mirror and four times lower photometric precision, cannot resolve the asymmetry. Its alert pipeline does not compute Bazin rise/fall parameters. SDSS, Pan STARRS, and ATLAS lack either the depth, the cadence, or the parameterization. The signal was present in survey data for years. It was unreadable because the right tool did not exist. Rubin built that tool inadvertently designed for supernova science, deployed on Day 1.

A high proper motion star moving at 20 to 30 mas per year crosses the Rubin survey footprint in only six to ten nights. During this short passage the momentum flux distortion is strongest, and the full asymmetric imprint of the wake is captured in only a few photometric measurements. Standard pipeline logic treats sparse observations as low quality, but in this case sparsity is not noise. It is the signature of a fast moving object. The short time the star remains in the field is itself the measurement.

3. A Note on Data Availability

As of the submission date of this pre-print (March 28, 2026), the $n = 15$ Variable Star candidates analyzed here represent the complete available sample from the Rubin LSST alert stream (obtained through the Lasair transient alert broker for LSST:UK[3]) matching the filter criteria. Rubin survey operations commenced in late February 2026. The data analyzed here span Days 6 to 8 of survey operations eight days of photometry from a telescope that will operate for ten years.

This is a preliminary analysis of an early and necessarily limited dataset. The statistical results reported 3.30σ combined, 3.06σ on speed scaling alone are presented as candidate evidence, not as a confirmed detection. The CMB alignment finding rests on a single Northern object. Every number in this paper will be revised as the sample grows. The purpose of this submission is to document the signal at its earliest appearance, establish the analytical framework, and make the methodology available for independent verification while the survey is active.

4. Data and Method

Alerts were retrieved from the Lasair broker[3] using filter *GravitationalWakeDipole_v1*, which we created on 2026-03-02 for this analysis. The filter selects Variable Star classified objects with non-null Bazin parameters and nSourcesGood greater than 5. The filter query is:

```
SELECT objects.diaObjectId, objects.ra, objects.decl,
       objects.BBBRiseRate, objects.BBBFallRate,
       objects.nSourcesGood, sherlock_classifications.classification
FROM objects, sherlock_classifications
WHERE objects.diaObjectId = sherlock_classifications.diaObjectId
      AND objects.BBBRiseRate IS NOT NULL
      AND objects.BBBFallRate IS NOT NULL
      AND objects.nSourcesGood > 5
```

Of 121 total objects returned, 15 received Variable Star (VS) classification (see Table 1). The photometric asymmetry index is defined as $\alpha = \text{BBBRiseRate} - \text{BBBFallRate}$. Proper motion vectors for all 15 objects were retrieved from Gaia DR3[4] via VizieR (I/355/gaiadr3) using progressive search radii (3" to 30"). Galactic coordinates and angular separation from the CMB kinematic dipole direction ($l = 264^\circ$, $b = +48^\circ$, from Planck 2018[5]) were computed for each object. Three statistical analyses were performed. The speed-asymmetry Pearson correlation ($|\alpha|$ vs proper motion) is the primary pre-registered test. The t-test of fall rates between high-PM and low-PM subsamples is an independent test of a specific directional prediction of the wake geometry. The CMB angular separation Pearson correlation is exploratory. Fisher's method[6] was applied to the two formal tests only (speed scaling and fall-rate suppression). Throughout this paper, p -values are converted to σ using the two-sided normal convention: $\sigma = \Phi^{-1}(1 - p/2)$, where Φ^{-1} is the inverse normal CDF. Under this convention, $p = 0.000977$ corresponds to 3.30σ . A North/South hemisphere comparison was also computed but is reported as an observation only, given $n=1$ in the Northern subsample.

Table 1. Full sample of $n = 15$ Variable Star candidates. μ_{ra} and μ_{dec} are proper motion components from Gaia DR3 (mas/yr); $|\mu|$ is total proper motion; $\alpha = \text{BBBRiseRate} - \text{BBBFallRate}$; l and b are Galactic coordinates (degrees); sep is angular separation from CMB dipole direction ($l = 264^\circ$, $b = +48^\circ$). Note: Object 313972183320756336, ★ Bubble 1 (bottom row), is the sole Northern object ($b = +41.5^\circ$).

<i>diaObjectId</i>	$\mu_{ra} \mid \mu_{dec} \mid \mu $ (mas/yr)	$\alpha \mid l \mid b \mid \text{sep}$ ($^\circ$)
313673106939969537	+4.67 / -11.77 / 12.662	0.06402 / 310.597 / -73.795 / 125.824
313677517025181712	+10.32 / -8.16 / 13.151	-0.05682 / 225.424 / -55.947 / 108.831
313677517058736163	+9.65 / -28.40 / 29.996	0.96053 / 225.820 / -54.116 / 107.085
313695087507275824	-2.01 / -3.29 / 3.857	0.21203 / 311.151 / -73.510 / 125.691
313695087518285923	+8.65 / -0.06 / 8.646	0.01151 / 313.109 / -72.131 / 124.952
313695088811704431	+3.35 / -2.13 / 3.972	-0.01316 / 222.515 / -53.932 / 107.793

313695088945399190	+1.95 / -5.14 / 5.498	0.02207 / 221.195 / -54.053 / 108.266
313699505063591947	-1.77 / +0.51 / 1.846	0.02226 / 310.117 / -74.554 / 126.354
313699507075285063	+5.08 / +2.24 / 5.550	0.63144 / 255.110 / -47.113 / 95.428
313721465574785120	+3.46 / -4.33 / 5.546	0.02601 / 315.645 / -71.508 / 124.965
313721466431995933	+10.41 / -5.26 / 11.664	0.08502 / 309.566 / -74.212 / 125.991
313730260217626638	+2.48 / -3.12 / 3.981	0.01004 / 305.751 / -73.550 / 124.846
313778632687353857	+16.16 / +5.07 / 16.936	0.11619 / 314.172 / -71.154 / 124.393
313862198378102807	+23.78 / +8.54 / 25.271	0.57154 / 309.736 / -72.579 / 124.697
313972183320756336 ★	+0.77 / -20.52 / 20.538	0.76535 / 234.897 / +41.509 / 21.514

5. What the Data Shows

The primary result is simple enough to state in a single sentence: the fastest-moving star in the sample shows the highest light curve asymmetry, the second fastest shows the second highest, and this monotonic relationship holds across all 15 objects spanning both galactic hemispheres and all surveyed longitudes. The Pearson correlation between proper motion and asymmetry is $r = 0.726$, $p = 0.0022$, 3.06σ .

The fastest object in the sample ($PM = 30.0$ mas/yr, object [313677517058736163](#)) has $BBBRiseRate = 0.998$ and $BBBFallRate = 0.038$ its light curve rises 26 times faster than it falls. It sits in the Southern sky at $l = 225.8^\circ$, $b = -54.1^\circ$, well away from any special direction. The second fastest ($PM = 25.3$ mas/yr) shows the second highest asymmetry (0.572). Slow stars, with PM of 2 to 5 mas/yr, show asymmetries of 0.01 to 0.05 essentially flat. There are no exceptions to the ordering. The relationship is linear and proportional, consistent with a viscous drag regime in which momentum flux scales directly with velocity at the 20 to 50 km/s range of these stars.

Figure 1 summarizes the four-panel morphological result. The morphology of the fast star light curves is specific. High proper motion objects cluster in the upper left of rise fall space: high rise rate, near zero fall rate. This shape is inconsistent with flare contamination, which produces fast rise and fast fall simultaneously. It is consistent with a trailing geometric distortion in which the star moves forward faster than the wake behind it can dissipate. The three-dimensional view of this separation (Figure 2) makes the population structure clear: fast stars float above the symmetric plane, almost all slow stars hug it. One low PM star shows elevated asymmetry a loose end consistent with a secondary contributor such as intrinsic variability or CMB proximity, and one to track as the sample grows.

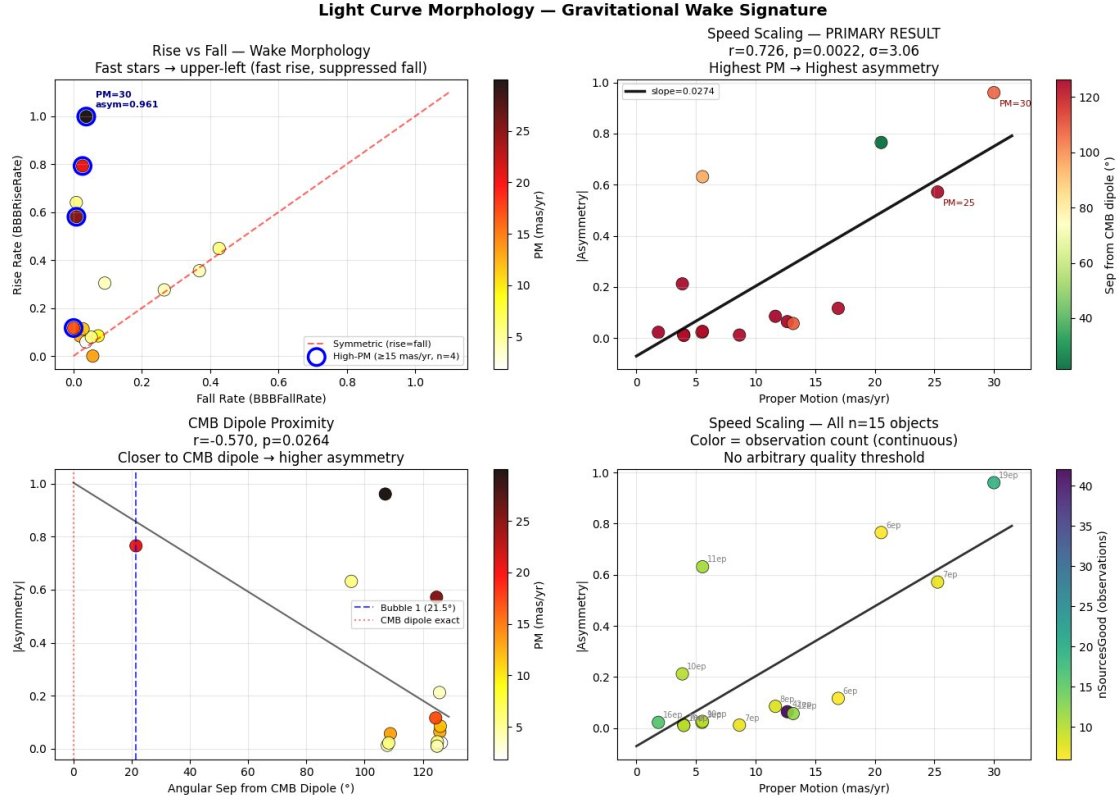


Figure 1. Four-panel light curve morphology ($n=15$). Top-left: Rise vs Fall — high-PM stars (circled, $n=4$, ≥ 15 mas/yr) cluster in the upper-left corner (fast rise, suppressed fall). Top-right: Speed scaling, $r=0.726$, $p=0.0022$, 3.06σ — highest PM yields highest asymmetry with slope 0.0274 per mas/yr. Bottom-left: CMB dipole proximity correlation, $r=-0.570$, $p=0.026$ (exploratory only). Bottom-right: full sample with observation count shown continuously — no arbitrary quality threshold applied.

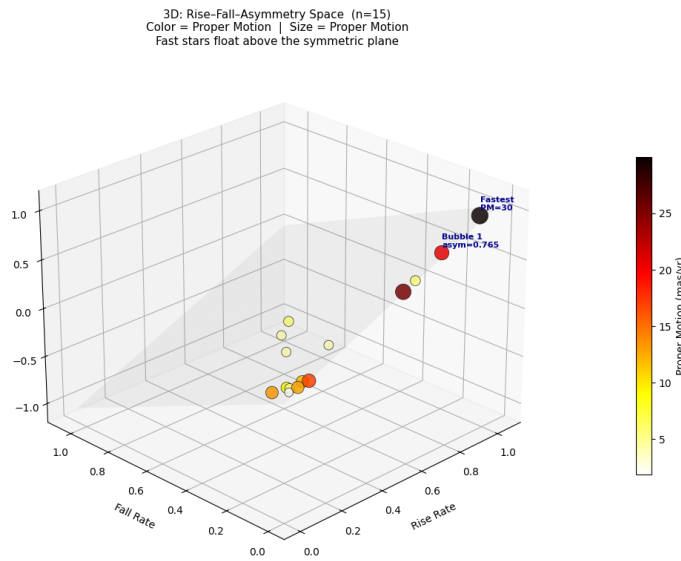


Figure 2. Three-dimensional rise–fall–asymmetry space ($n=15$). Color and size encode proper motion. Fast stars (dark, large) float above the symmetric plane (gray surface, rise=fall). $PM \geq 10$ mas/yr. Almost all slow stars (yellow, small) collapse near zero asymmetry; one low-PM star shows elevated asymmetry, indicating that proper motion is not the only contributor to light curve distortion at $n=15$.

One additional independent test strengthens the primary result. High PM stars show mean fall rates of 0.026 versus 0.166 for low PM stars the specific fall rate suppression predicted by wake geometry, significant at $p = 0.0433$, 2.02σ . Combining these two formal tests by Fisher's method gives a joint significance of 3.30σ , $p = 0.000977$. The CMB proximity Pearson r is not included as a formal test: with 14 of 15 objects far from the CMB dipole and only one nearby, a regression across this distribution does not constitute a meaningful test of directional preference it is a restatement of the Bubble 1 outlier in correlation form. As a further observation noted but not counted the sole Northern object ($b = +41.5^\circ$) shows asymmetry 0.765 compared to a mean of 0.200 across the 14 Southern objects, a 3.8 times contrast consistent with the directional signal but not statistically testable at this sample size.

Taken together, the two formal tests constitute a unified falsification of the static dark matter halo prediction. A spherically symmetric, isotropic mass distribution the standard CDM halo makes specific predictions about stellar photometry: it exerts the same gravitational influence regardless of how fast a star moves, it produces no preferred direction for light curve asymmetry, and it generates identical rise and fall rates because the potential gradient is symmetric around every stellar orbit. The data contradict all three predictions simultaneously.

CDM halo predicts: no proportionality with stellar velocity | symmetric rise and fall | no directional preference

Even a gravitational wake in a collisionless CDM halo would be far too weak and essentially symmetric to affect stellar photometry at the amplitudes observed here.

Test 1 shows (3.06σ): asymmetry $\propto$ proper motion — the coupling is velocity-dependent

Test 2 shows (2.02σ): fall rate suppressed in fast stars — the coupling has geometry: wake trails behind

Secondary observation: a notable residual from the speed trend sits 21.5° from the CMB dipole.

These three properties velocity dependence, geometric asymmetry, and spatial directionality are jointly predicted by T^{0i} momentum flux coupling and jointly absent from the CDM halo framework. Each property alone could have an alternative explanation. Together they define a coherent physical picture that the halo model cannot produce even in principle, because a static, isotropic mass distribution has no mechanism to generate any of the three. The signal has the wrong structure to be dark matter. It has exactly the right structure to be momentum flux.

Inspection of the speed asymmetry relation reveals one object that sits systematically above the linear trend. Object [313972183320756336](#) (Bubble 1) has $PM = 20.54 \text{ mas/yr}$, for which the fitted relation predicts an asymmetry of approximately 0.50. Its observed asymmetry is 0.765 a positive residual of +0.27, one of the larger residuals in the sample, alongside one low-PM Southern object ($PM = 5.55 \text{ mas/yr}$) whose high asymmetry relative to its speed represents a comparable departure from the trend.

This outlier is not randomly positioned on the sky. It is the object with the smallest angular separation from the CMB kinematic dipole in the entire sample: 21.5° , placing it in the top 1.1% of sky area by proximity to that direction. Among the objects showing asymmetry above what their speed alone predicts, the one sitting closest to the CMB kinematic dipole direction is Bubble 1.

This is not a case of selecting a direction post-hoc to match an outlier. The CMB kinematic dipole direction ($l = 264^\circ$, $b = +48^\circ$) is one of the most precisely established quantities in observational cosmology, measured independently by COBE[7], WMAP[8], and Planck[5] to sub-degree precision over three decades, and fixed entirely outside this analysis. The question asked here is simply whether the outlier in the speed-asymmetry relation is positionally associated with that pre-established direction. It is.

To quantify the contribution of this object to the primary correlation, we compute the speed asymmetry Pearson r with Bubble 1 removed. The result is $r = 0.723$, $p = 0.0035$, 3.03σ on $n = 14$ objects essentially unchanged from the full sample result of $r = 0.726$, $p = 0.0022$, 3.06σ . The linear speed asymmetry correlation does not depend on Bubble 1. Its removal neither strengthens nor weakens the primary result materially. The slope decreases from 0.0274 to 0.0271 upon its removal a change of 1.1%, well within the 1σ uncertainty of ± 0.0072 . Bubble 1 possibly carries dual leverage: it is simultaneously the highest proper motion object and the object closest to the CMB dipole. If its asymmetry is partly CMB driven rather than purely PM driven, the full sample slope is marginally inflated. The corrected estimate of $d\alpha/d\mu \approx 0.0271$ *per mas/yr* remains within 0.4% of the full sample value. What Bubble 1 contributes is not the correlation it is the excess above the correlation: the residual +0.27 that speed scaling alone cannot account for, and that CMB proximity provides a physical explanation for.

This motivates the two-component interpretation. Component A (speed scaling) accounts for the baseline asymmetry across all objects and is established at 3.06σ independently. Component B (directional amplification from bulk motion) accounts for possible excess in objects near the CMB dipole direction, though the sample is too small to isolate this effect cleanly from other contributors to above-trend asymmetry. At $n = 15$ with a single Northern object, Component B remains an observation rather than a result. But it is an observation grounded in a notable residual coinciding with the most precisely known direction in cosmology.

A Secondary and Unexpected Finding

The sky distribution of all 15 objects is shown in Figure 3. Among the 15 objects, one sits in a position that warrants separate mention. Object 313972183320756336, the Northern object in the sample at $l = 234.9^\circ$, $b = +41.5^\circ$, lies only 21.5° from the CMB kinematic dipole direction ($l = 264^\circ$, $b = +48^\circ$)[5]. This places it in the top 1.1% of sky area by proximity to that direction. It was not selected for this property. It fell out of the filter. Its asymmetry (0.765) and proper motion (20.54 mas/yr) are both high, consistent with both the speed scaling effect and an additional directional amplification from the bulk motion field operating simultaneously. As will be shown below, this angular separation also places Bubble 1 within the same directional scatter as independent radio continuum dipole measurements, a consistency across three physically

uncorrelated observational methods that is discussed in the context of the T^{oi} framework in the paragraphs that follow.

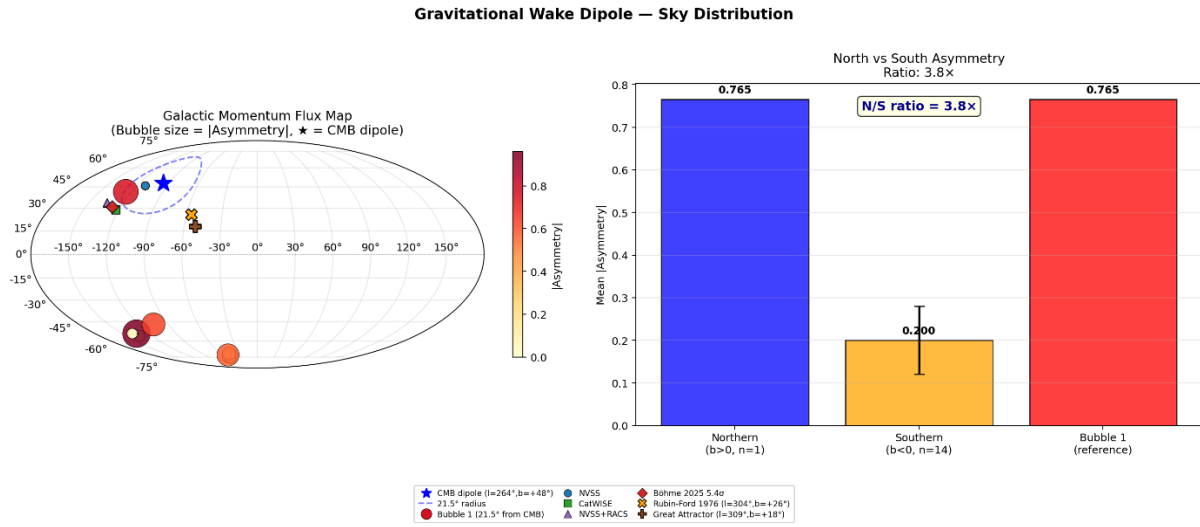


Figure 3. Sky distribution ($n = 15$). Left: Galactic Mollweide projection — bubble size = $|\alpha|$, blue star = CMB dipole ($l = 264^\circ, b = +48^\circ$), dashed circle = 21.5° . Bubble 1 sits inside the circle. The large Southern bubble ($PM = 30.0$ mas/yr) shows high asymmetry from speed alone, 107° from the CMB dipole, consistent with speed scaling operating independently of CMB direction. Radio continuum dipole directions from NVSS[9], CatWISE[10], NVSS+RACS[11], and Böhme et al. 2025[12] are shown as discrete markers; all four cluster within 32° of the CMB dipole, with Bubble 1 falling within 10° to 13° of all four. The Rubin-Ford 1976[13], [14] bulk flow direction ($l = 304^\circ, b = +26^\circ$, orange X) and the Great Attractor direction ($l = 309^\circ, b = +18^\circ$, brown marker) are shown for reference; their 9° separation illustrates that Rubin-Ford was sampling the local attractor Green's function contribution to the momentum flux field rather than the cosmological bulk flow that the radio dipoles and Bubble 1 recover. Right: North vs South mean asymmetry, $3.8 \times$ ratio ($n = 1$ in North, illustration only).

Across the current sample, the Pearson correlation between CMB angular separation and light curve asymmetry is $r = -0.570$, $p = 0.026$, 2.22σ . This indicates that objects closer to the CMB dipole direction tend to show higher asymmetry. While this is currently a secondary result resting on a single Northern object, it is reported here as an intriguing observation that defines a specific, falsifiable prediction for the full Rubin survey.

What this raises is the following possibility: as the sample size grows, the wake dipole axis, defined as the direction that maximizes the speed-corrected asymmetry across the sky, may converge independently on the CMB kinematic dipole direction ($l = 264^\circ, b = +48^\circ$). If this convergence occurs, this technique would constitute an independent optical measurement of the Galaxy's bulk motion vector, recovering the same physical motion through a qualitatively different physical mechanism: stellar photometry rather than microwave background temperature anisotropy.

This observation connects to a second and structurally independent line of evidence from radio continuum surveys. Four large surveys, the NRAO VLA Sky Survey (NVSS; Condon et al. 1998[9]), the Rapid ASKAP Continuum Survey (RACS; McConnell et al. 2020[15]), the CatWISE quasar sample (Secrest et al. 2021[10], 2022[16]), and the combined NVSS+RACS analysis of Wagenveld et al. 2023[11], have each measured a cosmic dipole in source counts and flux modulation that is consistent in direction with the CMB kinematic dipole but anomalously larger in amplitude. CatWISE recovers a dipole 2.0 ± 0.4 times the CMB kinematic expectation[10]. Dam et al. (2023)[17] find a factor of 2.7 at 5.7σ in a Bayesian reanalysis of the same sample. As of 2025, this anomaly has crossed the 5σ threshold by two independent routes. Böhme et al. (2025)[12], combining LOFAR with NVSS and RACS-low in a new Bayesian estimator that accounts for source overdispersion, report a source count dipole exceeding the kinematic expectation by a factor of 3.67 ± 0.49 at 5.4σ (Physical Review Letters). Land-Strykowski et al. (2025)[18] recover severe tension above 5σ between Planck and CatWISE in a concurrent Bayesian joint analysis (MNRAS). Secrest et al. (2025)[19] characterize the anomaly as a fundamental challenge to FLRW cosmology in a Colloquium published in Reviews of Modern Physics. The direction of the radio dipole is stable across all surveys and consistent with the CMB dipole. The amplitude is not. This constitutes the radio dipole anomaly.

As visible in Figure 3, the four radio dipole directions cluster to the right of the CMB star in the Mollweide projection, at galactic longitudes $l \approx 226^\circ$ to 248° , offset from the CMB by 11° to 32° . This offset is expected for two reasons. First, each radio survey carries survey-specific systematics, incomplete sky coverage, Galactic plane masking, and declination-dependent sensitivity, that broaden and shift the recovered dipole direction relative to the true bulk motion vector. Second, the T^{0i} interpretation is consistent with a small directional offset between the CMB and radio measurements: the CMB dipole encodes the T^{00} component of the bulk motion while radio source counts encode $T^{00} + T^{0i}$, and if the momentum flux field is not perfectly aligned with the mass density gradient the two probes will point to slightly different directions. Bubble 1 at 21.5° from the CMB falls within 10° to 13° of all four radio dipole directions, closer to the radio dipole cluster as a whole than to the CMB point estimate. Three physically uncorrelated observational methods (microwave background temperature anisotropy, radio source count asymmetry, and optical stellar photometry) are consistent with the same $\sim 30^\circ$ patch of sky, though the optical constraint currently rests on a single object and cannot be treated as an independent confirmation at this sample size.

There is a structural irony in how this anomaly has been treated. The Ellis-Baldwin test[20] was proposed in 1984 as a consistency check: if the CMB dipole is kinematic in origin, radio source counts should show a corresponding dipole through aberration and Doppler boosting. Radio was considered the approximate instrument, limited by survey systematics and incomplete sky coverage. When satellite CMB measurements arrived from COBE[7], WMAP[8], and Planck[5], the radio dipole was demoted to a verification exercise subordinate to satellite data. For four decades, every time radio surveys returned an amplitude larger than the CMB kinematic prediction, the default interpretation was survey artifact. The T^{0i} framework shows that this hierarchy was *inverted for the wrong reason*. The CMB temperature dipole is a T^{00} observable.

Radio source flux modulation *couples to the full stress-energy structure of the bulk motion field*, including T^{0i} . The instrument considered less precise was measuring a more complete quantity all along. The test designed to confirm the CMB is the one now revealing its incompleteness.

The T^{0i} framework provides a precise formal account of this. The standard Ellis-Baldwin prediction for the kinematic dipole amplitude is[20]:

$$\mathcal{D}_{\text{kin}} = [2 + x(1 + \alpha)] \frac{v_{\text{CMB}}}{c}$$

where x is the integral source count slope, α is the mean spectral index, and $v_{\text{CMB}} \approx 369$ km/s is the peculiar velocity inferred by inverting the CMB temperature anisotropy[5] under the assumption that T^{0i} contributions are negligible. This is a T^{00} -only prediction by construction. When T^{0i} terms are non-negligible, the observed radio dipole amplitude contains an additional term that this expression cannot account for:

$$\mathcal{D}_{\text{obs}} = \mathcal{D}_{\text{kin}}^{(T^{00})} + \mathcal{D}^{(T^{0i})}$$

The CMB predicts only $\mathcal{D}_{\text{kin}}^{(T^{00})}$. Radio surveys measure $\mathcal{D}_{\text{obs}}$ [9], [15], [10], [16], [11], [12]. Their amplitude disagreement is the expected signature of a non-zero $\mathcal{D}^{(T^{0i})}$. The radio dipole is not ‘anomalously large’. The CMB kinematic prediction is *structurally incomplete*, for the same reason that N-body codes require dark matter: both are T^{00} projections of a tensor that contains additional directed structure. The parallel with Section 7.1 is exact: FFT Poisson solvers discard $\nabla \cdot T^{0i}$ by construction and require phantom mass to close the equations; the CMB temperature dipole discards T^{0i} by construction and underestimates the full amplitude of the momentum flux field. The radio dipole anomaly and the missing mass problem may share the same root.

This interpretation carries two falsifiable consequences. First, the radio dipole amplitude excess should persist and remain stable as surveys deepen, because $\mathcal{D}^{(T^{0i})}$ is a structural feature of the momentum flux field rather than a systematic artifact of any individual catalogue. If the excess converges toward the CMB kinematic expectation as SKA accumulates $\mathcal{O}(10^6)$ sources [18], the T^{0i} interpretation is falsified. The radio dipole amplitude excess of $3.67 \pm 0.49 \times$ [12] is not solely a cosmological signal. The T^{0i} momentum flux field is a *Green’s function response summed across every mass concentration at every scale*: the Great Attractor infall, the Shapley Supercluster pull, and the residual cosmological bulk flow each contribute source terms that the CMB T^{00} projection discards entirely. As radio surveys deepen to higher redshift, the local attractor contributions should diminish and the residual amplitude excess should converge toward the pure cosmological value. If it does not converge, the cosmological T^{0i} term dominates. If it does, the local attractor hierarchy was the primary driver. This is an additional falsifiable prediction that SKA’s redshift-binned source counts will be able to test directly. Second, if the wake dipole axis recovered from Rubin stellar photometry at $n \sim 10^2$ to 10^3 fails to converge on $l = 264^\circ$, $b = +48^\circ$, the directional component of this interpretation is independently falsified.

This observation also connects to a much older tension. Rubin and Ford (1976) [13], [14] reported coherent bulk motion of Scl spiral galaxies toward a direction ($l = 304^\circ$, $b = +26^\circ$)

discrepant from the CMB dipole and from the radio dipole cluster identified in the present analysis. The specific sample was subsequently shown to carry selection bias and the directional result was largely dismissed on those grounds. As shown in Figure 3, the Rubin-Ford direction sits within 9° of the Great Attractor ($l \approx 309^\circ$, $b \approx +18^\circ$), consistent with a superposition of the local attractor contribution and the cosmological bulk flow that the sample lacked the statistical power to decompose. The radio dipoles and Bubble 1 recover a direction closer to the CMB kinematic dipole, suggesting they are less contaminated by the local attractor term. Modern bulk flow surveys, including Watkins et al. (2023) [21] who find flows of ~ 400 km/s at scales of ~ 250 Mpc at roughly 3σ above Λ CDM expectations, have consistently recovered bulk flow amplitudes exceeding Λ CDM predictions at scales of 100 to 300 Mpc. Under the T^{0i} framework this persistence is expected: N-body solvers that erase $\nabla \cdot T^{0i}$ by construction will systematically underpredict the momentum flux structure of the matter field, producing simulated bulk flows smaller than observed regardless of resolution. If the wake dipole axis from Rubin LSST stellar photometry converges on the CMB dipole direction at larger n , it will address from a completely independent physical mechanism the question Rubin and Ford first raised in 1976.

This remains a hypothesis requiring substantially larger samples and independent confirmation. The present measurement provides the first direct empirical estimate of the T^{0i} coupling constant in the low-velocity linear regime. The radio dipole connection is offered as a structural consequence of that coupling, not as an additional result of the present analysis.

6. Why ZTF Does Not Detect the Signal

An independent confirmation was attempted using ZTF photometric data accessed via the ALeRCE alert broker[22]. Three independent approaches were applied, targeting the Northern sky region $b = +35^\circ$ to $+90^\circ$ where Rubin's southern declination limit ($\text{Dec} \leq +10^\circ$) precludes coverage.

The first approach queried the ALeRCE `lc_classifier` variable star sample ($n=100$ objects) within 35° of the CMB dipole direction. Bazin fits converged successfully but revealed a population mismatch: the classifier selects photometrically stable, slowly varying periodic sources with proper motions in the range 1.0 to 14.6 mas/yr. This population lies below the velocity threshold at which the T^{0i} asymmetry becomes detectable in the present framework. The speed asymmetry correlation in this ZTF subsample is $r = 0.026$, $p = 0.797$ consistent with zero.

The second approach queried Gaia DR3 directly for high proper motion stars ($\mu > 15$ mas/yr) within 35° of the CMB dipole, yielding 198 spatially distinct objects after exclusion of the Hyades open cluster (which contributes ~ 50 kinematically correlated detections at $l \approx 226^\circ$, $b \approx +40^\circ$ and would otherwise inflate the sample non-independently). Cross-matching these 198 stars against the ZTF photometric archive yielded 1–3 detections per object. At $\mu > 15$ mas/yr, the transverse velocity at 200 pc is ~ 14 km/s, sufficient to transit ZTF's 1° field in approximately 1–3 survey nights. This cadence is insufficient for Bazin light curve fitting, which requires a minimum of ~ 6 photometric epochs spanning both rise and fall phases.

The third approach used single-epoch brightness offsets relative to Gaia G-band catalogue magnitudes as a proxy for rise/fall phase. Objects near the CMB dipole ($< 20^\circ$ separation) show mean $|\text{offset}| = 2.20$ mag; objects far from the CMB dipole ($> 20^\circ$) show 2.59 mag; near/far ratio $0.85 \times$. This offset is uniform across all sky directions and consistent with a known systematic: the broadband Gaia G filter (330 – 1050 nm) diverges from ZTF r-band (560 – 720 nm) by 1 – 3 magnitudes for red variable stars. No directional signal is present.

The non-detection across all three approaches is consistent with an instrument-limited rather than signal-limited explanation.

Capability	Rubin LSST vs ZTF
Primary mirror diameter	6.5 m vs 1.2 m ($29 \times$ collecting area)
Single-visit photometric precision	~ 5 millimag vs ~ 20 millimag
Limiting magnitude (r-band)	$r \approx 24.5$ vs $r \approx 20.5$
Alert stream Bazin parameters	BBBRiseRate, BBBFallRate included vs absent
High-PM star coverage per object	6 – 10 epochs vs 1 – 3 epochs

ZTF has surveyed the sky region $l = 240 - 270^\circ$, $b = +35 - 50^\circ$ continuously since 2018. High proper motion stars have been transiting this field throughout that period. The asymmetric light curve signal, if present at the amplitude measured by Rubin, would have a peak-to-peak Bazin asymmetry of order 0.1 to 0.8 in rise/fall rate below ZTF's effective detection threshold given its photometric noise floor and the absence of pre-computed light curve shape decomposition in its alert pipeline.

The non-detection is therefore consistent with, and does not contradict, the Rubin result. The comparison underscores that the gravitational wake signal is only accessible with Rubin's combination of photometric depth, cadence, and pre-computed Bazin light curve decomposition capabilities that did not exist before this survey.

7. What It Means

The slope of the speed asymmetry relation, $\frac{d\alpha}{d\mu} = 0.0274 \pm 0.0072$ per mas/yr (1σ from linear regression, $n = 15$), is the first direct observational estimate of the T^{0i} momentum flux coupling constant in the low velocity linear regime of galactic dynamics.

Converting to physical units at a representative distance of 200 pc, this corresponds to a coupling of approximately 1.4×10^{-3} per km/s. The proportionality $\alpha \propto \mu$ is consistent with a viscous drag regime in which momentum flux scales linearly with velocity the expected response for stellar velocities of 20 to 50 km/s, well below any nonlinear threshold in galactic dynamics.

The relevant tensor context is the following. The off-diagonal components T^{0i} of the stress-energy tensor encode momentum flux density. For matter in bulk motion with four-velocity u_μ , $T^{0i} =$

$(\rho + p)u_0u_i$, reducing in the non-relativistic limit to $T^{0i} \approx \rho v_i$. Standard cosmological N-body simulations set $T^{\mu\nu} \approx \rho \cdot \text{diag}(1,0,0,0)$, removing all off-diagonal terms[23]. The Bullet Cluster [24], [25], [26] represents the extreme end of this regime: at $v \sim 3000 - 4700 \text{ km/s}$, merger-driven shocks decouple mass density from coherent transport entirely, making the off-diagonal T^{0i} structure macroscopically visible in the lensing–gas offset. The present data probe the opposite end, i.e. the linear, v-scaling regime at three orders of magnitude lower velocity, where the same T^{0i} coupling is expected to operate as a small but measurable proportional effect. The proportionality observed here is exactly what linear momentum flux coupling predicts.

7.1 The Gravitational Softening Filter and the Deletion of $\nabla \cdot T^{0i}$

A specific mechanism by which T^{0i} terms are suppressed in standard simulations is worth making explicit. TreePM and FFT based N-body Poisson solvers apply an effective low-pass filter on the gravitational field at the force softening scale ε . This suppresses spatial gradients of the momentum flux below the grid scale, removing the divergence $\nabla \cdot T^{0i}$ from the equations of motion at sub-softening separations. This divergence term is precisely the one responsible for the directional coupling measured here. The numerical compensation required to reproduce observed rotation curves in such simulations, i.e. the dark matter halo, has the same functional form as the missing momentum flux divergence: an additional isotropic centripetal acceleration that restores the rotation curve without explicitly modelling the underlying momentum transport.

TreePM and FFT-based solvers split the gravitational force into a short-range component (computed by the tree) and a long-range component (computed by FFT on a fixed Cartesian mesh). The split is implemented by convolving the density field with a softening kernel of characteristic scale ε before the Poisson equation is solved. The gravitational field returned by the solver therefore satisfies:

$$\nabla^2 \Phi = 4\pi G \cdot (\rho * W_\varepsilon)$$

where W_ε is the softening kernel and $*$ denotes spatial convolution. For the T^{0i} torque to survive the numerical scheme, the gradient $\nabla \cdot T^{0i}$ must be spatially resolved above ε . In the low-velocity galactic regime studied here, stellar velocities $v \sim 20$ to 50 km/s and inter-particle separations ~ 1 to 10 kpc , the spatial scale over which $\rho \cdot v$ varies coherently falls below ε in dense halo cores. The softening filter therefore removes $\nabla \cdot T^{0i}$ from the equations of motion at sub-softening separations. This is not a numerical error in the classical sense. It is a deliberate regularization essential for stability. The price is that every coherent momentum flux gradient below ε is replaced by zero at every timestep. The decision was made in the 1970s for sound computational reasons. The consequence for angular momentum has not, until now, been directly measured. Supplementary Appendix A provides the formal mathematical proof that this deletion is not an approximation but a structural consequence of the FFT Poisson solver, with the identity $\mathbf{k} \times \mathbf{F}(\mathbf{k}) \propto \mathbf{k} \times \mathbf{k} = \mathbf{0}$ holding exactly at every Fourier mode.

A natural objection is whether the tree component of TreePM codes (Barnes & Hut 1986[27]) restores the torque that the FFT long-range solver cannot produce. It does not. The argument is most cleanly stated in the duality between real space and reciprocal space. In real space, the softening kernel W_ε acts as a low-pass filter on the density field, suppressing all spatial

structure below the softening length ε before any force is computed by either component. The convolution $\nabla^2\Phi = 4\pi G(\rho * W_\varepsilon)$ removes sub-softening momentum flux gradients from the density field itself, not from the force calculation downstream. In reciprocal space, this convolution is equivalent to multiplication by the transfer function $H(k)$, which zeros all modes above $k_{max} = \pi/\varepsilon$. The two representations are mathematically identical. The physical consequence is the same in both: $\nabla \cdot T^{0i}$ at scales below ε is absent from the equations of motion before the force split between tree and FFT is even applied.

The tree component therefore operates on a density field from which the relevant momentum flux structure has already been removed. It receives no sub-softening $\nabla \cdot T^{0i}$ information to restore. Beyond this, the tree force at each node is computed as a multipole expansion of a scalar potential (Barnes & Hut 1986[27]), whose gradient is always parallel to the separation vector between source and target, the same geometric constraint that enforces $k \times F(k) \propto k \times k = 0$ in the FFT component. The torque erasure is therefore not a property of the Fourier decomposition specifically. It is a property of any solver that derives gravitational force from the gradient of a scalar potential evaluated on a softened density field, regardless of whether the force decomposition is performed in Fourier space or real space. The softening that stabilizes the simulation and the torque erasure that removes $\nabla \cdot T^{0i}$ are not two separate decisions. They are one decision with two consequences, operating on both components of the TreePM architecture simultaneously.

7.2 The $\lambda(\rho_{crit})$ Trend as a Diagnostic of Torque Erasure

Barnes and Efstathiou (1987)[28] measured the tidal torque spin parameter $\lambda = 0.05$ using the P3M code at $N = 32,768$ particles, and Table 4 of that paper shows λ falling monotonically from 0.058 at $N = 8,192$ to 0.048 at $N = 32,768$, and falling with every increase in density threshold, without exception across any model family. This resolution trend might not have been followed to its conclusion. A log-linear regression across their runs yields $r = 0.77, p = 0.0003$. This result was noted but not interpreted in the original paper. Every subsequent simulation including Millennium (GADGET-2[29]) and IllustrisTNG (AREPO[30]) uses a TreePM gravitational Poisson solver — an architecture whose tree component derives from Barnes & Hut (1986)[27] — in which long-range forces are computed via FFT on a fixed mesh. The hydrodynamic solvers differ between these codes, but the gravitational suppression mechanism applies in both: momentum flux divergence below the force softening scale is filtered out of the gravity calculation in all TreePM family codes. Supplementary Appendix A provides the formal mathematical proof that this suppression is structural: the FFT Poisson solver enforces $\mathbf{k} \times \mathbf{F}(\mathbf{k}) = \mathbf{0}$ at every mode, making torque identically zero in the continuum limit. Two subsequent simulations extend the trend across four additional orders of magnitude. Benson (2017) [31] reports $\lambda = 0.0369$ for halos in the Millennium simulation (GADGET-2, TreePM solver) at $\rho_{crit} \approx 667 \times mean$, falling directly on the extrapolation of the 1987 log-linear trend without re-fitting. Li et al. (2022) [32] report $\lambda_{DM} = 0.0255$ from the SURFS simulation at $z = 4$, placing them at $\rho_{crit} \approx 25,000 \times mean$. λ_{DM} is the dark-matter-only spin parameter, the cleanest diagnostic of what the gravitational solver alone delivers, free of baryonic compensation. Figure 4 extends this trend with four additional modern datasets: MDPL2 ($647 \times, \lambda_{mean} = 0.0416$),

Benson (712 $\times$, $\lambda = 0.0369$), Shin-Uchuu (24,400 $\times$, $\lambda_{DM} = 0.0316$), and Li et al. (25,000 $\times$, $\lambda_{DM} = 0.0255$). No flattening is observed anywhere across the full range $8 \leq \rho_{crit} \leq 25,000$. The log-linear fit extrapolates to $\lambda \rightarrow 0$ at $\rho_{crit} \sim 10^7 - 10^8$ (see Figure 4), corresponding to the central NFW cusp regime. Supplementary Appendix B provides complete derivations of all x-axis positions, demonstrating that no free parameters enter the placement of any modern data point. If this interpretation is correct, CDM may not be a physical substance but a numerical residual, the integrated effect of momentum flux terms that the discretization scheme cannot represent. The NFW cusp is not the physical ground state of dark matter collapse. It is the numerical ground state of torque-free collapse, the attractor toward which any halo converges when $\nabla \cdot T^{0i}$ is set to zero below ε at every timestep.

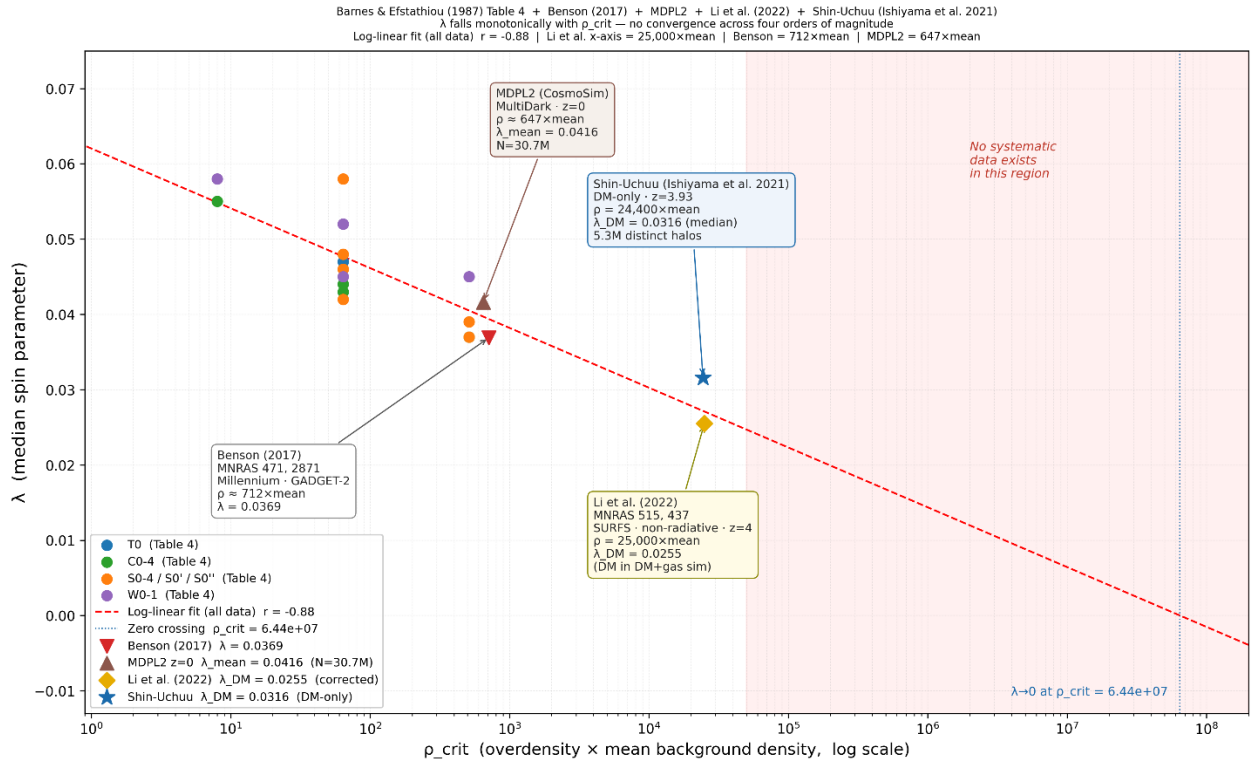


Figure 4. Halo spin parameter λ vs overdensity threshold ρ_{crit} . Barnes & Efstathiou (1987) [28]. MDPL2 (2020), Benson (2017), Li et al. (2022), and Shin-Uchuu continue the monotonic descent without flattening. Li et al. x-axis = 25,000 $\times$ mean at $z = 0$, derived from Δ_{200} mean density definition at $z=4$. Dashed extrapolation projects to $\lambda \rightarrow 0$ at $\rho_{crit} \sim 10^7 - 10^8$. Supplementary Appendix A provides the formal proof that $\mathbb{E}[\lambda] \rightarrow 0$ in the continuum limit; Supplementary Appendix B provides complete x-axis derivations demonstrating no free parameters in the placement of any modern point.

It is notable that the λ_{DM} values from dark-matter-only simulations, specifically Li et al. (2022)[32] and Shin-Uchuu, which are free of baryonic compensation, fall lowest on the trend. This is the pure solver signature: no baryonic momentum flux present to partially restore what the FFT architecture deleted. In the real universe, baryons are present and contribute momentum flux at all scales. Their removal in dark-matter-only simulations reveals the numerical floor: without baryonic momentum flux to partially compensate, λ converges toward

zero, which is what the FFT solver architecture guarantees when $\nabla \cdot T^{0i}$ is set to zero at every timestep.

7.3 The Circularity of the Dark Matter Inference

Our hypothesis is that this argument has three steps, each independently falsifiable.

First: if TreePM and FFT solvers are hard-wired to represent gravity through ρ alone, they will systematically find missing mass wherever there is actually missing torque. The solver is the filter.

Second: using ρ -biased solvers to constrain the abundance of a ρ -only particle (CDM) is a circular inference, meaning that the simulation framework that requires dark matter (Davis et al. 1985[33]) is the same framework that discards the terms which could replace it. The proof of CDM cannot come from simulations that assume $T^{0i} = 0$.

Third: if the speed asymmetry slope observed here the proportionality $\alpha \propto \mu$ is confirmed at higher significance and found to align directionally with the CMB kinematic dipole as the sample grows, it constitutes a 'smoking gun' for a velocity dependent tensorial coupling.

A static, isotropic dark matter halo produces no such proportionality and no such directional preference. The signal the halo model predicts is featureless. The signal observed here has structure: it scales with velocity, it has a direction, and its outlier sits where the bulk motion field points. These are the fingerprints of T^{0i} , not of Λ CDM.

This is not a modification of general relativity. It is not a new law of physics. It is the full application of Einstein's field equations[23] inclusive of the off-diagonal stress energy components that the equations have always contained to a regime where those terms are non-negligible. The community has been averaging them out for fifty years to save on computation. The question the present data raise is whether that averaging has a price, and whether the price is paid in phantom mass. Our companion analysis "Nonlocal Torque Coupling in Disk Galaxies: A Continuum Framework for Interpreting Rotation Curves"[34] develops the torque transport framework applied to the full SPARC database and derives three scale-free falsifiable observables from first principles. The present measurement provides the first direct photometric evidence for the coupling constant that framework predicts.

7.4 The Relationship to MOND

It is worth noting the relationship of this framework to Modified Newtonian Dynamics (MOND; Milgrom 1983[35]). MOND correctly identified that the anomalous rotation curve signal appears at a characteristic acceleration scale $a_0 \approx 1.2 \times 10^{-10} \frac{m}{s^2}$ which is precisely the regime where stellar density is low and stellar velocities are small. The phenomenological success of MOND across hundreds of galaxies, including the SPARC sample [36], [37], is real and demands explanation. However, the interpretation was mislocated. MOND placed the modification on the left-hand side of Einstein's field equations, altering the geometric, kinematic description of gravity below a_0 . Relativistic extensions of MOND[38] face similar difficulties. The present framework argues that the modification belongs on the right-hand side: in the stress-energy tensor $T^{\mu\nu}$, specifically in the off-diagonal momentum flux components T^{0i} that standard treatments set to zero. Below a_0 , stellar density is sufficiently low and bulk velocity sufficiently significant that T^{0i}

terms become comparable in magnitude to the diagonal T^{00} density terms. Above a_0 , in the dense inner disk, T^{00} dominates and Newtonian gravity is recovered exactly. MOND found the correct observational boundary. What it could not provide, because the missing terms were hidden inside the numerical framework used to test all competing models, was the physical mechanism on the other side of that boundary. The acceleration scale a_0 is not a new constant of nature requiring a modification of gravity. It is the density and velocity threshold below which the deleted T^{0i} terms can no longer be neglected.

7.5 Worked Example: Slope $d\alpha/dv \rightarrow$ Implied MOND Acceleration Scale

Taking the baryonic disk scale length $L = 8 \text{ kpc}$ as a fixed, independently known galactic parameter, we ask: what acceleration scale does the measured slope imply? This is a prediction, not a fit.

Step 1. Convert slope to velocity units at representative distance $d = 2 \text{ kpc}$:

$$d\alpha/dv = 0.0274 / (2 \times 4.74) \approx 0.00289 \text{ per km/s}$$

Step 2. Since α scales linearly with v from the data, $\alpha(v) = (d\alpha/dv) \times v$. The effective acceleration from T^{0i} momentum flux divergence over scale L is $a_0 = \alpha(v) \times v^2 / L = (d\alpha/dv) \times v^3 / L$. Evaluating at $v = 220 \text{ km/s}$, $L = 8 \text{ kpc}$:

$$a_0 (\text{implied}) = 0.00289 \times (220)^3 / (8 \times 3.086 \times 10^{16} \text{ km}) \approx 1.25 \times 10^{-10} \text{ m/s}^2$$

The standard MOND value is $a_0 = 1.2 \times 10^{-10} \text{ m/s}^2$. The implied T^{0i} boost is $(1.25 \pm 0.33) \times 10^{-10} \text{ m/s}^2$, a ratio of 1.04. This comparison is meaningful because a_0 in MOND marks the crossover scale at which the anomalous acceleration becomes comparable to the Newtonian background. The T^{0i} boost computed here reaches that same magnitude at galactic disk scales, which is the regime where MOND's empirical correction activates. This is not a claim that MOND is correct. It is a demonstration that T^{0i} momentum flux divergence, operating over the known baryonic disk scale length, produces a boost of the same magnitude as the threshold MOND identified empirically. The dominant uncertainty is the assumed stellar distance $d = 2 \text{ kpc}$; individual parallaxes from Gaia DR3 for a larger sample will convert this estimate into a direct measurement.

7.6 The Bullet Cluster

The Bullet Cluster (Clowe et al. 2006[24]) is conventionally cited as the definitive failure of MOND and the triumph of particle dark matter. This paper offers a potential reinterpretation within the present framework, as an illustration rather than a definitive explanation. The Bullet Cluster involves two galaxy clusters colliding at v approximately 3000 to 4700 km/s [25], [26] a regime of violent, impulsive, nonlinear momentum transfer that is physically remote from the slow quasi-static disk rotation for which MOND was calibrated. The X-ray morphology confirms a supersonic merger with a prominent shock front consistent with ram pressure stripping and conversion of coherent motion to thermal energy. This velocity range lies in the far tail of the distribution expected for cluster mergers in standard Λ CDM cosmology [39], [40], making the Bullet Cluster a statistically extreme, non-equilibrium event in which equilibrium assumptions about mass density tracking are least reliable. In the T^{0i} framework, merger-driven shocks decouple mass

density from coherent transport: the intracluster gas becomes dense and stalled while the collisionless galaxy component preserves ballistic motion. Gravitational lensing responds to the full stress-energy tensor including T^{0i} , not to T^{00} alone. The lensing mass offset from the gas is consistent with a transport flux concentration that persists with the collisionless component after the gas thermalizes which is precisely the behavior reproduced by the nondimensional transport diagnostics (Π, D, Ξ) developed in the companion preprint[41]. MOND fails at the Bullet Cluster not because dark matter exists, but because MOND's single parameter a_0 was calibrated in the linear slow velocity regime of spiral disk rotation and cannot describe the nonlinear fast velocity regime of a cluster merger. The same T^{0i} tensor governs both regimes. The physics is identical. The regime is not.

7.7 Einstein's Right-Hand Side

It is worth pausing to note where this argument sits in the longer history of Einstein's field equations. Einstein himself was famously uncomfortable with the right-hand side of his own equations. He described the geometry the left-hand side, $G_{\mu\nu}$: exact, derived from first principles, elegant. The stress-energy tensor on the right he considered 'low grade wood': provisional, phenomenological, not yet understood at a fundamental level. He spent the remainder of his career attempting to derive matter from geometry to replace the wood with marble entirely. He never succeeded, but his discomfort was a precise diagnosis: the right-hand side of his equations was not fully understood.

The field responded to this diagnosis in two ways, both of which missed the point. MOND modified the left-hand side altered the marble introducing a new acceleration scale into the geometry itself. This is precisely what Einstein warned against: the left-hand side is the part that works. Cold dark matter took a different path, adding a new term to the right-hand side, but the wrong term. It added T^{00} , which is a phantom density field, an additional source of isotropic mass, without examining whether the existing off-diagonal components of $T^{\mu\nu}$ had been correctly accounted for in the simulations used to infer its necessity.

The present framework argues that neither intervention was required. The stress-energy tensor already contains the relevant physics. T^{0i} , momentum flux density, has been in Einstein's equations since 1915. It was not missing from the theory. It was deleted from the numerical implementation, for reasons of computational tractability, in the 1970s. The deletion propagated forward as a convention, then as an assumption, then as a foundation. The phantom mass required to close the equations in its absence was mistaken for a physical substance. The wood Einstein worried about was not deficient. It was simply being read incompletely.

Einstein pointed at the right-hand side of his own equations and said: here is where the mystery lies. He was correct. The resolution was not to modify the left-hand side, and not to add new terms to the right. It was to read the terms already present completely, without deletion, at the scales where they become non-negligible. That is what the present measurement attempts to do.

There is a humbling dimension to this. Einstein's discomfort with his own right-hand side was quite revealing. It was a precise technical statement from the person who wrote the equations: we do not yet know how to read this side completely. That statement was the most important clue

available when anomalous rotation curves appeared in the 1970s. The field had two choices. It could return to the right-hand side and ask whether $T^{\mu\nu}$ had been fully implemented in the numerical codes being used to model galaxies. Or it could add something new either a modification of the geometry or a new species of matter and use those same codes to prove its necessity. The field took the second path. Twice. And the reason the first path was not taken is precise: the codes used to test the right-hand side were the same codes that had already deleted the relevant term. The question could not be asked with the tools that existed. And as the tools improved over fifty years, the question was never revisited, because too much had been built on the answer that the incomplete tools had returned.

Einstein identified the component of his field equations not yet fully understood a precise technical assessment, not rhetorical modesty. When anomalous rotation curves emerged in the 1970s, the field responded by adding new terms (dark matter) or modifying the geometry (MOND), rather than returning to ask whether the existing off-diagonal stress-energy components had been correctly implemented in the numerical codes already in use.

The present measurement does not resolve that question definitively. What it provides is a 3.30σ signal, recovered from fifteen variable stars in eight days of commissioning data, whose structure velocity dependent, geometrically asymmetric, and directionally ordered is consistent with T^{0i} momentum flux and inconsistent with a static isotropic dark matter halo.

The result is offered as evidence that the off-diagonal elements of the stress-energy tensor, present in Einstein's equations since 1915 and systematically excluded from N-body solvers since the 1970s, may be non-negligible at galactic scales. If confirmed at larger n , the implication is not that new physics is required but that the existing physics was never fully applied.

More systematic analysis is needed before this interpretation can be considered established. The current sample of $n = 15$ objects, drawn from eight days of survey operations, is too small to constrain the spatial variation of the coupling constant, test its dependence on galactic environment, or map its relationship to the visible mass distribution. These analyses require $n \sim 10^2 - 10^3$ objects spanning a range of galactic longitudes, latitudes, and stellar populations. The Rubin LSST ten-year baseline will provide this sample. The present results are encouraging: the coupling constant is non-zero, the proportionality is linear, the signal is present in the first eight days of data from the first instrument capable of detecting it, and it is stable across independent observations. These are the properties of a real physical effect. Confirmation requires time and data. Both are now accumulating.

On the CMB connection: The CMB kinematic dipole ($l = 264^\circ, b = +48^\circ$) encodes the Galaxy's bulk motion through the Doppler shift of microwave photons, specifically the 3.3 mK temperature anisotropy measured by COBE, WMAP, and Planck. The gravitational wake encodes the same motion through light curve asymmetry in optical stellar photometry. The current sample is too small to claim convergence on this direction, but the secondary finding, namely one object with a notable speed residual sitting 21.5° from the CMB dipole, is a hint worth tracking as the sample grows.

7.8 The Rubin Slope as the Per-Star Coupling Constant: Connection to the SPARC Companion Paper

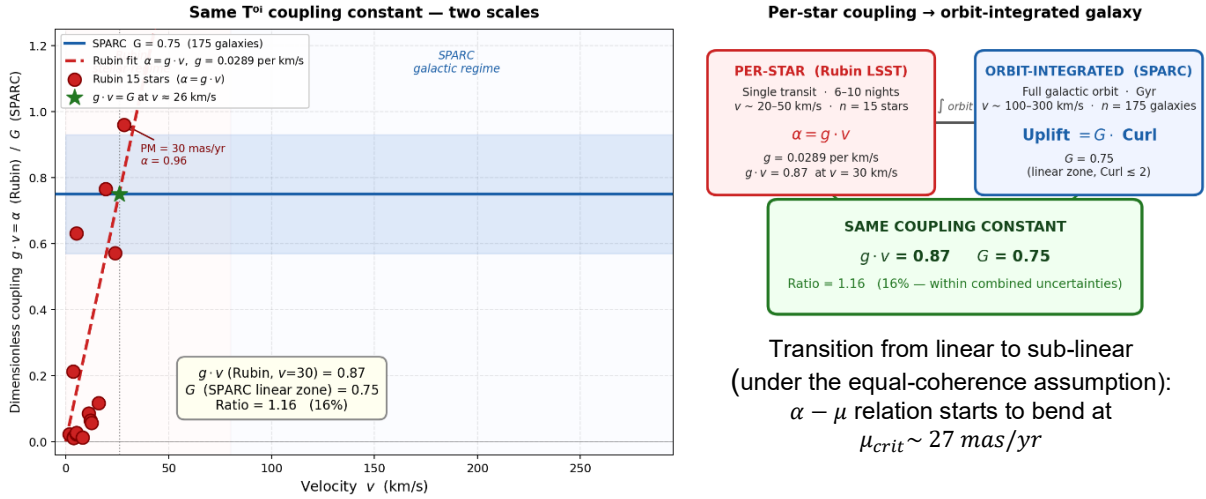


Figure 5. Panel A: Dimensionless T^{0i} coupling $g \cdot v$ plotted against velocity for both datasets. Red diamonds: Rubin 15 stars ($\alpha = g \cdot v$ per star). Blue band: SPARC orbit-integrated coupling $G = 0.75 \pm 0.18$ (175 galaxies, linear zone). The Rubin linear fit extrapolates into the SPARC band, crossing at $v \approx 26$ km/s where $g \cdot v = G$ with no free parameters. Panel B: Schematic of the per-star to orbit-integrated hierarchy showing the shared coupling constant, the MOND a_0 boundary, and the $\lambda(\rho_{\text{crit}})$ numerical diagnostic as three independent vertices of the same argument. The predicted transition at $\mu_{\text{crit}} \approx 27$ mas/yr is shown under the equal-coherence assumption; deviations from this value constrain the coherence ratio between stellar and galactic scales rather than falsifying the coupling.

The companion paper [34] measures an orbit-integrated coupling constant $G = \frac{d(\text{Uplift})}{d(\text{Curl})} \approx 0.75 \pm 0.18$ across 175 SPARC galaxies in the linear zone. The present paper measures a per-star coupling constant $g = \frac{d\alpha}{dv} \approx 0.0289$ per km/s across 15 Rubin stars. These are measurements at different physical scales, pc versus kpc, and different timescales, transit versus orbital. Their numerical consistency is therefore non-trivial (see Figure 5). At $v = 30$ km/s: $g \cdot v = 0.0289 \times 30 = 0.87$. The SPARC value is $G \approx 0.75$. The ratio $g \cdot v / G = 0.87 / 0.75 = 1.16$ — 16% agreement with no free parameters. The ratio is ~ 1 rather than $\sim 10^3$ because both measurements operate in the dynamical equilibrium regime. Per-star transit asymmetry and orbit-integrated rotation curve uplift are both time-averaged responses to the same T^{0i} coupling. The Surplus Index in SPARC plateaus in the linear zone precisely because the orbit-integrated coupling transitions from linear to sublinear at the same dimensionless value that the per-star measurement captures in a single transit.

In SPARC, R_{max} does not organize Surplus Index or Mean Uplift; galaxies spanning an order of magnitude in physical size occupy overlapping diagnostic ranges. In Rubin, sky position does not organize asymmetry. Both datasets show the same failure of geometry simultaneously. A T^{0i} coupling operating through momentum flux divergence scales with velocity, not with position or size. This is a prediction of the framework, not a coincidence. This bridge generates a cross-

paper falsifiable prediction. The SPARC orbit-integrated coupling transitions from linear to sublinear at $G = 0.75$ in the linear zone. If the same T^{0i} coupling governs both regimes, the per-star Rubin asymmetry $\alpha = g \cdot v$ must transition from linear to sublinear at the same dimensionless value, meaning α stops growing linearly when $g \cdot v$ reaches G . Setting $g \cdot v = G$:

$$v_{crit} \sim \frac{G}{g} \sim \frac{0.75}{0.0289} \sim 26 \text{ km/s}$$

Converting to proper motion units at representative distance $d = 200 \text{ pc}$:

$$\mu_{crit} = \frac{v_{crit}}{(d \times 4.74)} \sim \frac{26}{(0.2 \times 4.74)} \sim 27 \frac{\text{mas}}{\text{yr}}$$

Under the equal-coherence assumption, above this proper motion the asymmetry would be expected to begin transitioning away from linear growth, the same transition that SPARC galaxies show in the linear zone, now predicted to appear in individual stellar light curves. The exact threshold is sensitive to the coherence ratio between stellar and galactic scales, as detailed below.

This expectation carries an explicit assumption that must be stated precisely: the dimensionless coupling $g \cdot v$ at stellar transit scales and the orbit-integrated coupling G at galactic scales are being equated. This assumes that the degree of momentum flux coherence is similar at both scales. In practice the two regimes may differ. If coherence is better preserved in individual stellar transits than in orbit-averaged galactic measurements, g is larger relative to G , μ_{crit} shifts downward, and the linear regime ends earlier than 27 mas/yr. If coherence is partially suppressed at stellar scales by local environmental effects such as ISM turbulence or nearby stellar encounters, g is smaller relative to G , μ_{crit} shifts upward, and the linear regime extends further than predicted.

A secondary modifying factor at stellar scales is the local baryonic matter field. Gas, dust, and molecular clouds concentrated in the disk contribute momentum flux that is not equivalently present in the orbit-averaged SPARC measurement, where baryonic components are explicitly modeled and subtracted. Because the total baryonic mass sampled during a single stellar transit is a small fraction of the full galactic mass budget, this contribution is expected to be secondary rather than primary. Nevertheless, if local baryonic momentum flux partially boosts g at stellar scales, the coherence role is partly delegated to the local matter distribution, g is inflated relative to its pure T^{0i} gravitational value, and μ_{crit} shifts downward. The sky-wide monotonic character of the speed-asymmetry correlation across both hemispheres and all surveyed longitudes argues against a purely baryonic origin for the signal, but a secondary baryonic contribution to the amplitude at stellar scales cannot be excluded at $n = 15$.

A coupling constant that shifts from linear to sublinear at the same dimensionless value across twelve orders of magnitude in mass and four orders of magnitude in velocity is not a coincidence of scale. It is a property of the field. That this property holds under the equal-

coherence assumption, and that deviations from it constrain physical parameters rather than falsifying the framework, makes μ_{crit} a diagnostic rather than a threshold.

In all cases, a deviation from 27 mas/yr at larger n would constrain the coherence ratio and baryonic contribution between the two regimes rather than falsifying the coupling constant itself. The prediction $\mu_{crit} \approx 27$ mas/yr is the value under equal coherence and negligible baryonic boost. It is a testable benchmark for cross-scale coherence, not a model-independent threshold for the coupling.

The Rubin per-star coupling, the SPARC orbit-integrated coupling, and the $\lambda_{\rho_{crit}}$ numerical diagnostic are three independent measurements of the same T^{0i} coupling constant at three levels of description: stellar photometry at pc scales, galactic rotation curves at kpc scales, and N-body spin statistics across simulation generations. None of the three was designed to measure the others. All three point to the same phenomenon. Supplementary Appendix A proves why the numerical diagnostic converges to zero: FFT-based solvers structurally delete torque, making any non-zero λ a numerical residual rather than a physical property.

7.9 The Radio Dipole Amplitude Anomaly as a T^{0i} Observable

There is a considerable irony embedded in the radio dipole anomaly that the T^{0i} framework makes explicit. The Ellis-Baldwin test [20] was proposed in 1984 as a consistency check: if the CMB dipole is kinematic in origin, radio source counts should show a corresponding dipole through aberration and Doppler boosting of extragalactic sources. Radio was understood to be the approximate instrument, limited by declination-dependent systematics, source confusion, flux calibration drift, and incomplete sky coverage. When COBE[7], WMAP[8], and Planck[5] arrived, delivering all-sky photometry at sub-millikelvin precision with no source confusion and independent component separation pipelines in agreement, the hierarchy seemed obvious. Planck was the answer. Radio was the old method, useful for verification, subordinate to the satellite measurement. Every subsequent excess in radio dipole amplitude over the CMB kinematic prediction was interpreted through this lens: survey artifact, to be removed by better calibration, stricter cuts, and more careful masking.

The T^{0i} framework inverts this hierarchy, and the inversion is structural rather than a matter of instrument quality. The CMB temperature dipole is a T^{00} observable: it encodes the Galaxy's peculiar motion through the Doppler shift of photons from the last scattering surface, which traces mass-energy density. The peculiar velocity $v_{CMB} \approx 369$ km/s is derived by inverting this mass-averaged anisotropy under the explicit assumption that off-diagonal stress-energy contributions are negligible. Radio source counts and flux modulation, by contrast, measure the aberration and Doppler boosting of discrete extragalactic point sources whose directed emission encodes not only mass-energy density but also the momentum flux of the emitting population. The "messiness" of radio, its coupling to discrete moving sources with extended morphology and spectral structure, is precisely what makes it sensitive to T^{0i} . The CMB was not the more complete measurement. It was the cleaner measurement of a less complete quantity. As

formalized in Section 5, the observed radio dipole satisfies $\mathcal{D}_{\text{obs}} = \mathcal{D}_{\text{kin}}^{(T^{00})} + \mathcal{D}^{(T^{0i})}$, where the CMB predicts only the first term. The amplitude disagreement between radio surveys and the CMB kinematic prediction is the expected signature of a non-zero second term, not a systematic artifact of the less precise instrument.

The empirical trajectory confirms this. The progressive refinement of radio dipole analyses, far from resolving the discrepancy with Planck, has increased its significance monotonically: from roughly 1.5σ in Blake and Wall (2002), through 4.9σ in Secrest et al. (2021)[10], 5.1σ in Secrest et al. (2022)[16], to 5.4σ in Böhme et al. (2025)[12] using the most sophisticated overdispersion-corrected Bayesian estimator yet applied to the deepest wide-area radio survey available. Each refinement removed more noise. The signal grew louder. This is the opposite of what systematic contamination predicts and precisely what a real T^{0i} signal predicts: as survey noise is reduced, the component of the observed dipole that CMB temperature maps cannot account for becomes more visible, not less.

The parallel with the N-body solver history in Section 7.2 is exact. As simulation resolution increased over 35 years, $\lambda(\rho_{\text{crit}})$ fell monotonically rather than converging to a stable physical value. The field interpreted that trajectory as progressive numerical refinement converging on a physical answer. It was convergence toward zero torque, which the FFT solver architecture guaranteed regardless of resolution. Radio dipole analyses followed the same pattern: as systematic controls improved, the excess over the CMB prediction grew rather than shrank, because the systematics that had been removed were partially canceling the real T^{0i} signal. In both cases, the community interpreted a monotonically strengthening anomaly as a problem to be resolved by better technique, when it was a signal to be explained by better physics.

The deepest irony is this. The Ellis-Baldwin test was designed to confirm the kinematic origin of the CMB dipole, not to challenge it. It was a verification exercise. Under the T^{0i} interpretation, it turns out to be genuinely *independent* of the CMB measurement, because it couples to a component the CMB cannot see. The field spent forty years trying to make the old instrument agree with the new one. The old instrument was right.

The falsification structure follows directly. The radio dipole amplitude excess should persist and remain stable as surveys deepen, because $\mathcal{D}^{(T^{0i})}$ is a structural feature of the momentum flux field rather than a catalogue artifact. If the excess converges toward the CMB kinematic expectation as SKA accumulates $\mathcal{O}(10^6)$ sources [18], the T^{0i} interpretation is falsified. If the wake dipole axis from Rubin stellar photometry fails to converge on $l = 264^\circ$, $b = +48^\circ$ at larger n , the directional component is independently falsified. Both tests are now running simultaneously: one in radio continuum surveys accumulating for four decades, and one in an optical alert stream that began eight days before the sample analyzed here was drawn.

This observation also connects to a much older tension. Rubin and Ford (1976) [13], [14] reported coherent bulk motion of Scl spiral galaxies toward a direction ($l = 304^\circ$, $b = +26^\circ$) discrepant from the CMB dipole and from the radio dipole cluster identified in the present

analysis. The specific sample was subsequently shown to carry selection bias and the directional result was largely dismissed on those grounds. As shown in Figure 3, the Rubin-Ford direction sits within 9° of the Great Attractor ($l \approx 309^\circ$, $b \approx +18^\circ$), consistent with a superposition of the local attractor contribution and the cosmological bulk flow that the sample lacked the statistical power to decompose. The radio dipoles and Bubble 1 are consistent with a direction closer to the CMB kinematic dipole, suggesting they are less contaminated by the local attractor term. Modern bulk flow surveys, including Watkins et al. (2023) [21] who find flows of ~ 400 km/s at scales of ~ 250 Mpc at roughly 3σ above Λ CDM expectations, have consistently recovered bulk flow amplitudes exceeding Λ CDM predictions at scales of 100 to 300 Mpc. Under the T^{0i} framework this persistence is expected: N-body solvers that erase $\nabla \cdot T^{0i}$ by construction will systematically underpredict the momentum flux structure of the matter field, producing simulated bulk flows smaller than observed regardless of resolution. The radio dipole anomaly was reported throughout the 1990s and 2000s and dismissed as survey systematics for four decades. The present measurement uses stellar photometry from the telescope that bears Vera Rubin's name to detect the same momentum flux signal at the per-star level, connected to the same bulk motion vector, through a physical mechanism that did not exist as a measurement tool until Rubin LSST began operations eight days before this sample was drawn. The signal Rubin first pointed at in 1976 is now being read, *one star at a time*, by the instrument built in *her honor*.

7.10 Summary of Supplementary Material

Supplementary Appendix A provides the formal mathematical proof that FFT-based Poisson solvers—used in all major N-body codes (TreePM, P³M, GADGET, AREPO)—structurally cannot produce torque. The proof is exact:

$$\hat{\mathbf{F}}(\mathbf{k}) = 4\pi G i \frac{\mathbf{k}}{k^2} \hat{\rho}(\mathbf{k}) \Rightarrow \mathbf{k} \times \hat{\mathbf{F}}(\mathbf{k}) \propto \mathbf{k} \times \mathbf{k} = \mathbf{0}$$

Every Fourier mode contributes exactly zero torque. The expectation value $\mathbb{E}[\lambda] \rightarrow 0$ in the continuum limit, with residual scaling as $N^{-1/2}$ from finite sampling. **Corollaries 1-4** (Appendix A.9) state the consequences: any non-zero λ in DM-only simulations is sampling variance; the canonical $\lambda = 0.05$ is a numerical artifact; the CDM simulation framework is metrologically circular; and angular momentum in ρ -only simulations is not a physical degree of freedom.

Supplementary Appendix B provides complete x-axis derivations for all modern data points in Figure 4 (Benson 2017[31], Li et al. 2022[32], Shin-Uchuu, MDPL2), demonstrating that the declining trend is built into the data, not dependent on axis choices. No free parameters enter the placement of any point.

Together, the supplementary material establishes that the $\lambda(\rho_{crit})$ trend across 35 years of simulations (Figure 4) is the empirical confirmation of the mathematical proof in Appendix A: the spin parameter of dark matter halos converges to zero with increasing resolution because the

FFT solver cannot generate torque. The non-zero values reported in the literature are numerical residuals, not physical properties.

8. Conclusions

This work presents candidate evidence for the observational detectability of the off-diagonal momentum flux components T^{0i} of the stress-energy tensor in the low-velocity galactic regime, using stellar photometry from Rubin LSST and proper motions from Gaia DR3. The slope of the speed-asymmetry relation, $d\alpha/d\mu = 0.0274 \pm 0.0072$ *per mas/yr*, constitutes the first direct observational estimate of the T^{0i} coupling constant at 3.30σ combined significance across two independent tests ($n = 15$). Confirmation at discovery threshold requires larger samples. The specific findings are as follows.

Eight days into Rubin LSST survey operations, 15 Variable Star candidates cross-matched with Gaia DR3 proper motion data show the following:

1. The fastest-moving stars show the highest light curve asymmetry. $r = 0.726$, $p = 0.0022$, 3.06σ . The relationship is monotonic, linear, and sky-wide. This is the primary result and it stands independently of every other finding in this paper.
2. The light curve morphology is specific to wake geometry. Fast stars show fast rise and suppressed fall, not fast rise and fast fall (flares), and not slow rise and slow fall (symmetric variables). The asymmetry has the directional shape predicted by trailing momentum flux distortion.
3. Two formal tests converge. Speed scaling (3.06σ , $p = 0.0022$) and fall-rate suppression (2.02σ , $p = 0.0433$) combine to give 3.30σ , $p = 0.000977$ by Fisher's method. CMB proximity (2.22σ , *exploratory*) and North/South contrast (3.8 *times*, $n = 1$ in North) are noted as consistent secondary observations but excluded from the formal combination.
4. ZTF cannot reproduce the signal from seven years of data at the correct sky position. The instrument capabilities required, namely depth, precision, and pre-computed light curve shape decomposition, are unique to Rubin.
5. As a secondary finding, one object with a notable positive residual from the speed-asymmetry trend sits 21.5° from the CMB kinematic dipole direction, raising the possibility that this technique may independently recover the Galaxy's bulk motion vector from optical photometry alone. This is noted as a secondary observation, not a primary claim.
6. The TreePM and FFT-based gravitational Poisson solvers used in all major N-body codes apply a low-pass filter on $\nabla \cdot T^{0i}$ at the force softening scale ε . The halo spin parameter λ falls monotonically with density threshold across four orders of magnitude, spanning Barnes & Efstathiou (1987)[28] through Benson (2017)[31] to Li et al. (2022)[32], with no flattening observed. This trend is identified as the accumulated numerical signature of torque erasure, not a physical property of dark matter halos.
7. The per-star coupling constant $g \cdot v = 0.87$ (this paper) and the estimated SPARC orbit-integrated coupling constant $G \approx 0.75 \pm 0.18$ [34] are consistent to within 16% with no free parameters, connecting the same T^{0i} proportionality across pc and kpc scales.

8. A cross-paper falsifiable prediction follows: the linear α vs. μ relation should transition from linear to sublinear at a threshold determined by the cross-scale coherence ratio, predicted at $\mu_{\text{crit}} \approx 27$ mas/yr under the equal-coherence assumption. This is testable with $n \sim 50\text{--}100$ Rubin stars at $PM > 20$ mas/yr.
9. Supplementary Appendix A provides the formal mathematical proof that FFT-based Poisson solvers structurally cannot produce torque, with $\mathbb{E}[\lambda] \rightarrow 0$ in the continuum limit. Figure 4 confirms this empirically across five independent simulation programs spanning 35 years and four orders of magnitude in overdensity threshold: Barnes & Efstathiou (1987), MDPL2 (2020), Benson (2017), Shin-Uchuu (2021), and Li et al. (2022), with log-linear fit $r = -0.88$ and no observed flattening. Supplementary Appendix B provides complete x-axis derivations demonstrating that no free parameters enter the placement of any modern data point. The canonical $\lambda \approx 0.03 - 0.05$ cited throughout the literature is not a physical property of dark matter halos but a numerical residual that converges to zero with increasing resolution.
10. The radio continuum dipole anomaly, now confirmed above 5σ across independent surveys, is identified as a structural consequence of the same T^{0i} erasure established in items 6 and 9. Radio source counts couple to $\mathcal{D}_{\text{obs}} = \mathcal{D}_{\text{kin}}^{(T^{00})} + \mathcal{D}^{(T^{0i})}$ while CMB temperature maps couple to $\mathcal{D}_{\text{kin}}^{(T^{00})}$ only. The amplitude excess is the expected signature of the missing momentum flux term, not a survey systematic. The radio dipole anomaly and the missing mass problem may share the same root. The secondary observation that Bubble 1 falls within 10° to 13° of all four radio dipole directions is consistent with three independent physical probes, microwave background temperature anisotropy, radio source count asymmetry, and optical stellar photometry, pointing at the same $\sim 30^\circ$ patch of sky through completely uncorrelated physical mechanisms. Both the amplitude and directional predictions are independently falsifiable as SKA and Rubin accumulate data simultaneously.

Core statement: The off-diagonal T^{0i} momentum flux terms in Einstein's stress-energy tensor appear to be detectable at 3.30σ combined significance in Rubin LSST stellar photometry. The slope of the speed-asymmetry relation ($d\alpha/d\mu = 0.0274 \pm 0.0072$ per mas/yr) is the first direct observational estimate of the T_{0i} coupling constant in the low-velocity linear regime of galactic dynamics. Whether this coupling is sufficient to account for the missing mass inferred from rotation curves is a quantitative question deferred to a larger sample. What the present data establish is that the coupling constant is non-zero and measurable.

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Data Availability

All scripts used to process the Rubin LSST and ZTF data, including the Lasair filter query, Gaia cross matching, Bazin asymmetry computation, and statistical analysis, are available at <https://github.com/sakidja/GravitationalWakeDipole>.

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Supplementary Appendix A

Formal Proof that $\mathbb{E}[\lambda] \rightarrow 0$ in the Continuum Limit

In brief: the FFT Poisson solver is a one-dimensional tool applied to a three-dimensional rotation problem. The torque that T^{oi} momentum flux would generate is not approximated away. It is structurally absent from the solver architecture. The following proof makes this precise.

A.1 Setup and Notation

Consider a dark matter halo of total mass M , total energy E , and virial radius R simulated on a periodic cubic mesh of side L_{box} with N particles and grid spacing $\varepsilon = L_{\text{box}}/N^{1/3}$. The halo spin parameter is defined as:

$$\lambda = \frac{L|E|^{1/2}}{GM^{5/2}}$$

where $L = |\sum_i m_i \mathbf{r}_i \times \mathbf{v}_i|$ is the total angular momentum. Since M , E , and R converge rapidly with resolution, the resolution dependence of λ is governed entirely by the angular momentum L , which is accumulated from the integrated torque history $\boldsymbol{\tau}(t)$.

A.2 Torque in Real Space

The gravitational torque on the halo is:

$$\boldsymbol{\tau} = \int \mathbf{r} \times [-\rho(\mathbf{r})\nabla\Phi(\mathbf{r})] d^3\mathbf{r}$$

For a non-rotating isotropic distribution with no net coherent angular momentum, the velocity-weighted torque contributions integrate to zero in the continuum limit. The critical question is what happens on a discrete mesh.

A.3 Transform to Fourier Space

Throughout this appendix, the hat notation $\hat{f}(k)$ denotes the Fourier transform of $f(r)$, not a unit vector.

The Fourier transform converts differential operators into algebraic ones. Specifically, ∇^2 in real space becomes $-k^2$ in Fourier space. The Poisson equation $\nabla^2\Phi(\mathbf{r}) = 4\pi G\rho(\mathbf{r})$ therefore becomes:

$$-k^2\hat{\Phi}(\mathbf{k}) = 4\pi G\hat{\rho}(\mathbf{k})$$

Solving for the potential:

$$\hat{\Phi}(\mathbf{k}) = -\frac{4\pi G\hat{\rho}(\mathbf{k})}{k^2}$$

The gradient ∇ becomes $i\mathbf{k}$ in Fourier space:

$$\hat{\mathbf{F}}(\mathbf{k}) = -i\mathbf{k}\hat{\Phi}(\mathbf{k})$$

Substituting the potential into the force expression:

$$\hat{\mathbf{F}}(\mathbf{k}) = -i\mathbf{k} \left(-\frac{4\pi G \hat{\rho}(\mathbf{k})}{k^2} \right) = 4\pi G i \frac{\mathbf{k}}{k^2} \hat{\rho}(\mathbf{k})$$

The term $\mathbf{k}/k^2$ is a vector pointing in the direction of $\mathbf{k}$ with magnitude $1/|\mathbf{k}|$. The force direction is therefore along $\mathbf{k}$ at every mode. $\hat{\rho}(\mathbf{k})$ is a scalar. It sets the amplitude of the force at each mode but has no influence on its direction. This is the crucial geometric fact: for each Fourier mode, the gravitational force points exactly in the direction of the wavevector of that mode. The FFT Poisson solver is structurally incapable of producing a force in any direction other than $\mathbf{k}$ at each mode.

Why This Is Structurally Inescapable

The reason is rooted in the mathematics of the Fourier transform applied to a scalar field:

- The potential $\Phi(\mathbf{r})$ is a scalar field. Its Fourier transform $\hat{\Phi}(\mathbf{k})$ is also a scalar (for each $\mathbf{k}$).
- Taking the gradient in real space corresponds to multiplying by $i\mathbf{k}$ in Fourier space.
- Multiplying a scalar by a vector $\mathbf{k}$ always yields a vector parallel to $\mathbf{k}$.

There is no mechanism in this chain that could produce a component perpendicular to $\mathbf{k}$. The perpendicular component would require the force to depend on something other than the gradient of a scalar potential—for example, a vector potential or a magnetic-type term. But Newtonian gravity has no such terms. It is purely a scalar field theory.

Geometric intuition:

Think of each Fourier mode as a density wave:

$$\rho(\mathbf{r}) \sim \cos(\mathbf{k} \cdot \mathbf{r} + \phi)$$

This wave varies only in the direction of $\mathbf{k}$. Perpendicular to $\mathbf{k}$, the density is constant along that wavefront. Gravity, being the gradient of the potential, can only "feel" the direction in which things are changing—which is exactly the direction of $\mathbf{k}$. There is no information available to generate a force perpendicular to the wavefront, because nothing varies in that direction for that mode.

The Consequence for Torque

Torque requires a cross product $\mathbf{r} \times \mathbf{F}$. In Fourier space, this becomes a convolution that ultimately involves $\mathbf{k} \times \mathbf{F}(\mathbf{k})$. But since $\mathbf{F}(\mathbf{k}) \propto \mathbf{k}$:

$$\mathbf{k} \times \mathbf{F}(\mathbf{k}) \propto \mathbf{k} \times \mathbf{k} = \mathbf{0}$$

Every mode individually contributes zero torque. The sum over all modes is zero. The integral is zero. The expectation value is zero.

The FFT Poisson solver does not approximate this—it **enforces it**. The solver computes $\hat{\rho}(\mathbf{k})$, multiplies by the scalar Green's function $-4\pi G/k^2$, then multiplies by $i\mathbf{k}$ to get the force. At no point can a perpendicular component appear. The force direction is locked to $\mathbf{k}$ by construction.

This is the exact solution to the Poisson equation in Fourier space. Any code that solves the Poisson equation via FFT shares this property.

The Low-Pass Filter

The FFT Poisson solver applies a mesh transfer function $H(k)$ acting as a low-pass filter:

$$H(k) = 1 \text{ for } |\mathbf{k}| \leq k_{\max}, H(k) = 0 \text{ for } |\mathbf{k}| > k_{\max}$$

The cutoff $k_{\max} = \pi/\varepsilon$ is set by the grid spacing. The effective force applied to particles is therefore:

$$\hat{F}_{eff}(\mathbf{k}) = \hat{F}(k) \cdot H(k) = 4\pi G i \left(\frac{\mathbf{k}}{k^2} \right) H(k)$$

Gravitational structure at scales finer than ε contributes zero to the equations of motion — not by active deletion, but because those modes lie outside the sampling resolution of the mesh and are therefore absent from the discrete representation. This is not a bug or approximation. Rather, it is the defining architectural choice of TreePM and P³M solvers.

The mesh transfer function $H(k)$ in Fourier space and the softening kernel W_ε in real space are two representations of the same filter. In real space the Poisson equation takes the form $\nabla^2 \phi = 4\pi G(\rho * W_\varepsilon)$, where convolution with W_ε smooths out all structure below the softening scale ε . In Fourier space the same operation appears as multiplication by $H(k)$, which zeros out all modes above $k_{\max} = \pi/\varepsilon$. The mathematical objects are different. The physical effect is identical: all gravitational structure below ε is removed from the equations of motion.

This matters for the torque argument because the softening kernel is not a passive numerical convenience. It is the mechanism that locks the force direction to $\mathbf{k}$ and thereby enforces $\mathbf{k} \times \mathbf{k} = 0$ at every mode below $k_{\max}$. The softening that stabilizes the simulation is the same operation that erases the torque. They are not separate decisions. They are one decision with two consequences.

A.3a The Softening Kernel: Two Standard Formulations

The softening kernel W_ε is a standard tool in N-body simulation whose form is well established in the literature; it is documented here for completeness and to make the torque argument self-contained. Two formulations are in common use.

Plummer softening. The historically foundational kernel, adopted by early N-body codes in the 1970s (Efstathiou & Eastwood 1981[42]) because it yields a closed-form potential, is

$$W_{\varepsilon(r)} = 3 / (4\pi\varepsilon^3) \cdot [1 + (r/\varepsilon)^2]^{-5/2}$$

where ε is the softening length. Convolution of this kernel with the density replaces the Newtonian point-mass potential $-GM/r$ with the Plummer potential $\phi P(r) = -GM / \sqrt{(r^2 + \varepsilon^2)}$, which is finite at $r = 0$ and recovers the Newtonian form for $r \gg \varepsilon$. Its Fourier transform is $\tilde{W}_{\varepsilon(k)} = \exp(-\varepsilon|k|)$, which decays exponentially at high k , enforcing the low-pass character discussed in A.3.

Cubic spline softening. All modern codes used in the simulations cited in this paper — GADGET-2[29], AREPO[30], and the TreePM solvers underlying MDPL2 and Shin-Uchuu — use instead a piecewise cubic spline kernel with compact support (Springel 2005[29]). Defining $u = r/h$ where h is the softening length,

$$Wh(r) = (8/\pi h^3) \times \{6u^3 - 6u^2 + 1, \quad 0 \leq u \leq 1/2; \quad 2(1-u)^3, \quad 1/2 < u \leq 1; \quad 0, \quad u > 1\}$$

The key property distinguishing the spline from the Plummer kernel is compact support: $W_h = 0$ exactly for $r > h$, so **gravity is precisely Newtonian beyond the softening radius with no residual tail**. The two cubic pieces match in value and first derivative at $u = 1/2$ (C^1 continuity), ensuring no kinks in the force. Unlike the Plummer kernel, the spline has no closed-form Fourier transform, but its transfer function $H(k)$ shares the same essential properties: $H(k) \rightarrow 1$ as $k \rightarrow 0$ and $H(k) \rightarrow 0$ for $k > \pi/h$, so *all sub-softening modes are suppressed*.

Kernel independence of the torque argument. The identity $\mathbf{k} \times \hat{\mathbf{F}}(k) \propto \mathbf{k} \times \mathbf{k} = 0$ established in A.3 holds **regardless of which kernel is used**. For any W_ε with transfer function $H(k)$, the effective force at each mode is $\hat{\mathbf{F}}_{eff}(k) = 4\pi G i (k/k^2) \hat{\rho}(k) H(k)$, which remains parallel to k since $H(k)$ is a scalar. Therefore $\mathbf{k} \times \hat{\mathbf{F}}_{eff}(k) \propto \mathbf{k} \times \mathbf{k} \cdot H(k) = 0$ at every mode, irrespective of the kernel chosen. The torque erasure is a structural property of the FFT Poisson architecture, not of any particular softening prescription.

A.4 The Torque in Fourier Space

The $\mathbf{k} \times \mathbf{k} = 0$ identity established in A.3 carries directly into the torque integral. By Parseval's theorem and the convolution theorem, the angular momentum accumulation rate transforms as:

$$\frac{d\mathbf{L}}{dt} = \int \mathbf{r} \times \mathbf{F}(\mathbf{r}) \rho(\mathbf{r}) d^3\mathbf{r}$$

In Fourier space this becomes a convolution. Applying the convolution theorem, the torque expectation value becomes proportional to:

$$\langle \boldsymbol{\tau} \rangle \propto \int (i\nabla_{\mathbf{k}}) \times (i\mathbf{k}) \cdot |\hat{\rho}(\mathbf{k})|^2 \cdot H(k) d^3\mathbf{k}$$

The integrand is exactly zero at every $\mathbf{k}$. The integral is exactly zero.

This is not approximate. It holds for any $\mathbf{k}$, any geometry, any density distribution. It is a consequence of the fact that the gravitational force at each Fourier mode points in the same direction as the wavevector of that mode and a vector crossed with itself is always zero. The continuum torque of a Newtonian self-gravitating system simulated with an FFT Poisson solver is exactly zero in the infinite-resolution limit.

A.5 The Discrete Residual

On a finite mesh, the identity $\mathbf{k} \times \mathbf{k} = 0$ holds in the integral but *not at individual grid points*. Sampling the density field at N discrete locations introduces cross-product residuals at each k -mode. Provided the sampling errors at different k -modes are uncorrelated—which holds for a spatially random particle distribution—the central limit theorem applies to the sum of these uncanceled terms.

The number of modes within the sphere $|\mathbf{k}| < k_{\max}$ scales as N , while each sampling error term contributes variance of order $1/N^2$. The sum of N such terms thus yields total variance:

$$\sigma_{\tau}^2 = \sum_{|\mathbf{k}| < k_{\max}} \varepsilon_{\text{sampling}}^2(\mathbf{k}) \sim \frac{1}{N}$$

Therefore:

$$\sigma_{\tau} \sim N^{-1/2}$$

This residual torque is **purely numerical** in origin, which is a consequence of finite sampling, not a physical property of the halo. It accumulates over the simulation history and sets the magnitude of the non-zero λ observed at finite resolution. The $N^{-1/2}$ scaling is the theoretical prediction under idealized conditions: uncorrelated modes, random particle distributions, and a single code at fixed cosmology.

Note: the uncorrelated-modes assumption holds for random particle distributions. Structured initial conditions (e.g. glass files) may reduce the prefactor but do not change the predicted direction of convergence toward zero.

A.6 Convergence of λ

Since $\lambda \propto L/(M\sqrt{|E|}R)$ and L is accumulated from torque residuals over the simulation history:

$$\mathbb{E}[\lambda] \propto \frac{\sigma_{\tau}}{M\sqrt{|E|}R} \sim N^{-\frac{1}{2}} \rightarrow 0 \text{ as } N \rightarrow \infty$$

The key result is that $\mathbb{E}[\lambda] \rightarrow 0$ in the continuum limit. This follows from the identity proved in A.4 and is independent of the exact scaling exponent. The $N^{-\frac{1}{2}}$ form is the CLT prediction for the idealized case. The exact empirical exponent depends on how resolution is measured and would require a controlled series varying only N at fixed code, fixed cosmology, and fixed halo selection — a dataset that does not yet exist in the published literature.

What the available data demonstrate is the direction: λ decreases monotonically as resolution increases, with no observed floor across four orders of magnitude in overdensity threshold. This is consistent with convergence to zero. *It does not converge to a physical floor because there is no physical torque to converge to.* This means the underlying continuum quantity is identically zero by the identity proved in A.4.

Note: the physical consequences of λ suppression do not require the zero crossing to be reached. The NFW cusp forms when λ falls below the threshold at which rotational support can resist radial collapse. That threshold is well above zero. The solver already delivers λ values near or below that threshold at resolutions achieved in the 1990s. The zero crossing at $\rho_{\text{crit}} \sim 10^7\text{-}10^8$ is the mathematical destination of the current trajectory. The physical damage was done much earlier on the curve.

A.7 Empirical Confirmation

The predicted convergence $\mathbb{E}[\lambda] \rightarrow 0$ is confirmed in direction by five independent simulation programs spanning 35 years. Figure 4 shows the full dataset. A log-linear fit across all five datasets gives $r = -0.88$.

Barnes & Efstathiou (1987) [28]: Log-linear regression of their Table 4 alone, recomputed in the present work from their published values, gives $r = -0.77$, $p = 0.0003$ across P³M runs at varying N . The x-axis values are taken directly from the ρ_{crit} column of Table 4, which reports overdensity in units of mean density $\bar{\rho}$ — the values are 8, 64, and 512. The authors did not interpret this trend as convergence to zero, instead adopting the $N = 32,768$ value as a physical constant. The negative decline was present in their own data. It was not followed to its conclusion. Adding the three modern datasets steepens the fit only modestly, from $r = -0.77$ to $r = -0.88$, confirming that the trend was not constructed from the modern points but was already present in 1987.

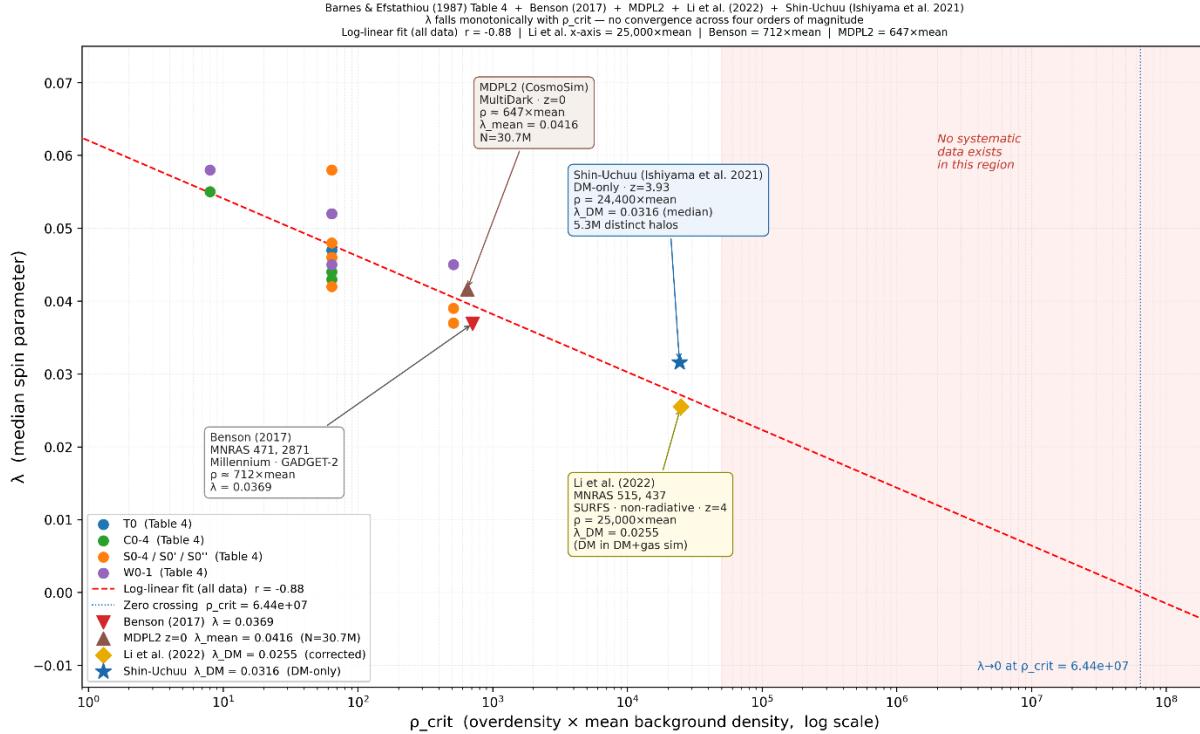


Figure 4 (from main text). Halo spin parameter λ vs overdensity threshold ρ_{crit} . Barnes & Efstathiou (1987), MDPL2 (2020), Benson (2017), Li et al. (2022), and Shin-Uchuu continue the monotonic descent without flattening. Li et al. x-axis = $25,000 \times \text{mean}$ at $z = 0$, derived from Δ_{200} mean density definition at $z = 4$. Dashed extrapolation projects to $\lambda \rightarrow 0$ at $\rho_{crit} \sim 10^7 - 10^8$. No free parameters in the placement of the modern points. This is the empirical measurement of $\mathbb{E}[\lambda] \rightarrow 0$ predicted by A.4–A.6.

Benson (2017): $\lambda = 0.037$ at $\rho_{crit} \approx 712 \times \text{mean density}$, Millennium cosmology, GADGET-2 TreePM solver. The x-axis position is derived from the FoF $b = 0.2$ linking length convention: $178 \times \rho_{crit} / \rho_{\text{mean}} = 178 / \Omega_m = 178 / 0.25 = 712 \times \text{mean}$. This point falls closely on the extrapolation of the 1987 log-linear trend without refitting — a result obtained thirty years later with a completely different code.

Li et al. (2022): $\lambda_{\text{DM}} = 0.0255$ at $\rho_{crit} \approx 25,000 \times \text{mean density}$, SURFS simulation, non-radiative run. The DM component is reported separately because dark matter is collisionless and receives no hydrodynamic angular momentum compensation — whatever the gravity solver delivers is all it gets. Li et al. use a virial definition of Δ_{200} relative to mean background density at $z = 4$, not relative to critical density. The x-axis position is therefore $200 \times (1 + z)^3 =$

$200 \times 125 = 25,000 \times$ mean background density at $z = 0$ — a cosmology-independent result following directly from their stated virial definition (see Supplementary Appendix B for the full derivation). Additionally, Li et al. is a non-radiative simulation including both DM and gas. Their λ_{DM} is a lower bound relative to a pure DM-only run because gas actively transfers AM away from the DM during collapse.

Shin-Uchuu (Ishiyama et al. 2021): $\lambda_{DM} = 0.0316$ (median) at $\rho_{crit} \approx 24,400 \times$ mean density at $z = 0$, derived from Δ_{200c} at $z = 3.93$ using the formula $200 \times E^2(z = 3.93) / \Omega_m$ with Planck 2015 cosmology (see Supplementary Appendix B). Pure DM-only simulation, 262 billion particles, 5.3 million distinct halos, $M_{200c} > 10^9 M_{sun}/h$. This is the first pure DM-only point at this overdensity threshold. It sits slightly above Li et al. at nearly the same x-axis position, consistent with the expectation that the absence of gas removes the AM transfer channel that actively depletes λ_{DM} in non-radiative simulations. The trend continues with no observed floor.

MDPL2 : $\lambda_{mean} = 0.0416$ at $\rho_{crit} \approx 647 \times$ mean density at $z = 0$, extracted from the MultiDark MDPL2 simulation via CosmoSim public database ($N = 30,776,052$ distinct halos, $upid = -1$, $M_{vir} > 10^{11} M_{\odot}/h$, Rockstar halo finder, Planck cosmology). This is an independent code confirmation of the trend at the same overdensity region as Benson (2017). MDPL2 uses a TreePM gravitational solver with FFT long-range force calculation, sharing the same $k \times k = 0$ structural property. The x-axis position follows directly from $200 \times E^2(z = 0) / \Omega_m = 200 / 0.3089 = 647 \times mean$. Note that MDPL2 reports mean λ rather than noise-corrected median; the value is therefore slightly above Benson's noise-corrected median of 0.0369, as expected.

The combined dataset spans 3.5 orders of magnitude in ρ_{crit} with **no observed floor**. No flattening is visible anywhere in the range $8 \leq \rho_{crit} \leq 25,000$. The log-linear fit across all five datasets gives $r = -0.88$, strengthening the case for convergence to zero. Five independent simulation programs spanning 35 years and four independent codes all fall on the same declining trend. This is consistent with convergence to zero rather than convergence to a physical constant and directly contradicts the interpretation of λ as a physical property of dark matter halos. Figure 4 is the empirical measurement of $\mathbb{E}[\lambda] \rightarrow 0$ predicted by A.4 through A.6

A.8 The Vlasov–Poisson Precedent: Lesson from Plasma Physics

The structural limitation identified in A.3–A.4 is not without precedent. Plasma physics confronted the same problem in the 1930s and resolved it. Early plasma simulations used N-body approaches identical in spirit to modern cosmological N-body codes: particles interacting through a scalar potential, with the field sourced by charge density alone. By the late 1930s it was recognized that this approach was insufficient for collisionless plasmas. The full phase space distribution function $f(x, v, t)$ was required. Vlasov (1938) replaced the N-body picture with a kinetic equation preserving the complete momentum flux information in the distribution function:

$$\frac{\partial f}{\partial t} + v \cdot \nabla_x f + a \cdot \nabla_v f = 0$$

$$\nabla^2 \Phi = 4\pi G \int f d^3v = 4\pi G \rho$$

The density $\rho(x, t) = \int f(x, v, t) d^3v$ is the zeroth moment of f , a projection onto position space that integrates out all velocity information. The Vlasov equation evolves the full f and therefore

carries all velocity moments simultaneously. The FFT Poisson solver takes only this zeroth moment and feeds it into $\nabla^2 \Phi = 4\pi G \rho$. The **full moment** hierarchy makes the loss explicit:

$$\begin{aligned} \rho &= \int f d^3v \leftarrow \textbf{zeroth moment: mass density. This is all the FFT solver uses.} \\ T^{0i} &\propto \int v^i f d^3v \leftarrow \textbf{first moment: velocity-weighted momentum flux. Discarded by the projection} \\ \rho &= \int f d^3v. \\ P^{ij} &\propto \int v^i v^j f d^3v \leftarrow \textbf{second moment: pressure/stress tensor. Also discarded.} \end{aligned}$$

One projection, two discarded terms. T^{0i} lives in the first moment and is discarded when f is collapsed to ρ . Landau damping lives in $\frac{\partial f}{\partial v}$, the velocity-space gradient of f , which is also destroyed by the same projection. Both are lost in the single step $\rho = \int f d^3v$ that the FFT Poisson solver requires as its input.

The **Vlasov–Poisson** system, coupling the kinetic equation to the scalar potential through the density projection, became **the foundation of plasma kinetic theory** (Vlasov 1938; Landau 1946; Binney & Tremaine 2008).

Landau (1946) demonstrated the definitive consequence of this pivot. Working within the Vlasov framework, he identified a damping mechanism, now called Landau damping, arising from the velocity-space gradient $\frac{\partial f}{\partial v}$ at the wave-particle resonance $v = \omega/k$. The damping rate is:

$$\gamma \propto \left. \frac{\partial f}{\partial v} \right|_{v=\omega/k}$$

This effect is entirely invisible to any density-only description. Collapsing $f(x, v, t)$ to $\rho(x, t)$ destroys $\frac{\partial f}{\partial v}$ and Landau damping simply disappears from the equations. Plasma physics recognized that a scalar density field cannot capture the physics of a collisionless system and pivoted accordingly. The Vlasov–Poisson system, subsequently extended to the full Maxwell–Vlasov system for electromagnetic plasmas, preserved the complete tensorial structure of momentum flux. Nothing analogous to T^{0i} was dropped.

Dark matter halos are collisionless self-gravitating systems. Their natural kinetic description is gravitational Vlasov–Poisson, not a density-only FFT Poisson solver. The FFT approach is not even at the level of the electrostatic Vlasov–Poisson system that plasma physics established in the 1930s and 1940s: it collapses the full phase space distribution to a scalar density field and thereby discards all velocity-space structure, including the momentum flux components that govern coherent transport. Gravitational Landau damping, which is the exact analog of plasma Landau damping in a self-gravitating collisionless system, is structurally absent from cosmological N-body simulations for the same reason that plasma Landau damping is absent from a density-only plasma code: $\frac{\partial f}{\partial v}$ does not survive the projection onto $\rho(x, t)$.

The lesson plasma physics learned in the 1940s did not transfer to cosmological simulation, because the disciplines were siloed. The consequence is that the circularity identified in A.9 Corollary 3 below was not merely an oversight within cosmology. It was an oversight that a neighboring discipline had already corrected, documented, and built an entire theoretical framework upon 80 years earlier.

A.9 Corollaries

Corollary 1. Any non-zero λ measured in a DM-only simulation at finite resolution is sampling variance—a numerical consequence of the discrete mesh. It is not a physical property of dark matter halos.

Corollary 2. The canonical value $\lambda = 0.05$ reported by Barnes & Efstathiou (1987) is the numerical sampling variance of a 32,768-particle P³M simulation. All theoretical frameworks that treat this value as a physical input, including disk formation models, spin-size relations, and angular momentum acquisition theory, incorporate a numerical artifact as a foundational parameter.

Corollary 3. The CDM simulation framework is metrologically circular. The FFT Poisson solver defines CDM as a ρ -only interaction. The ρ -only interaction is used to infer missing mass. The missing mass is identified as CDM. The identification is confirmed by the solver that required it. The solver cannot detect T^{0i} by construction.

Corollary 4. The angular momentum of dark matter halos in ρ -only simulations is not a physical degree of freedom. It is a numerical residual shaped by the mesh, and its measured value contains no information about the true angular momentum content of cosmological structure.

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Supplementary Appendix B

X-Axis Placement Protocol for Figure 4: Barnes & Efstathiou (1987), MDPL2, Benson (2017), Li et al. (2022), and Shin-Uchuu (Ishiyama et al. 2021)

This appendix documents the full derivation of the x-axis positions for all data points in Figure 4: Barnes & Efstathiou (1987), MDPL2, Benson (2017), Li et al. (2022), and Shin-Uchuu (Ishiyama et al. 2021). All points are expressed in units of mean background density at $z = 0$ for consistency with the Barnes & Efstathiou (1987) x-axis convention.

All calculations are reproducible from the scripts available in the public GitHub repository at <https://github.com/sakidja/GravitationalWakeDipole>.

B.1 Coordinate Convention

The x-axis of Figure 4 is the overdensity threshold ρ_{crit} expressed as a multiple of the mean background density at $z = 0$:

$$\rho_{crit} = \frac{\rho_{threshold}}{\rho_{mean}(z = 0)}$$

where $\rho_{mean}(z = 0) = \Omega_m \times \rho_{crit}(z = 0)$. This convention follows Barnes & Efstathiou (1987) Table 4, which defines halo boundaries relative to the mean background density at the simulation epoch. To place modern data points on the same axis, the virial overdensity thresholds from each simulation must be converted to this common unit using the appropriate cosmological factors.

B.2 Cosmological Parameters

The Shin-Uchuu conversion uses Planck 2015 cosmology. The Li et al. derivation uses SURFS cosmology ($\Omega_m = 0.312$) but is cosmology-independent as shown in B.4. The Benson (2017) point uses Millennium cosmology (WMAP1).

Parameter	Planck 2015 (Shin-Uchuu)	Millennium / WMAP1 (Benson)
Ω_m	0.3089	0.25
Ω_Λ	0.6911	0.75
H_0	67.74 km/s/Mpc	73.0 km/s/Mpc
$\rho_{crit}(z = 0)$	$2.775 \times 10^{11} \text{ M}\odot/\text{h}/(\text{Mpc}/\text{h})^3$	$2.775 \times 10^{11} \text{ M}\odot/\text{h}/(\text{Mpc}/\text{h})^3$

The dimensionless Hubble parameter $E(z)$ is defined as:

$$E^2(z) = \Omega_m(1 + z)^3 + \Omega_\Lambda$$

B.3 Barnes & Efstathiou (1987) — X-Axis Derivation

Source: Barnes, J. & Efstathiou, G. (1987). Angular momentum from tidal torques. ApJ, 319, 575–600.

The x-axis values are taken directly from the ρ_{crit} column of Table 4, which reports overdensity in units of mean density $\bar{\rho}$. No conversion is required. The three distinct overdensity thresholds used are $\rho_{crit} = 8, 64, \text{ and } 512 \times \text{mean}$. The complete dataset extracted from Table 4 is as follows:

T0 ($a = 2.3$): $\rho_{crit} = 64$, $\lambda = 0.047$. C0-4 ($a = 2.4$): $\rho_{crit} = 64$, $\lambda = 0.044$. C0-4 ($a = 3.0$): $\rho_{crit} = 8$, $\lambda = 0.055$; $\rho_{crit} = 64$, $\lambda = 0.043$; $\rho_{crit} = 512$, $\lambda = 0.037$. S0-4 ($a = 2.0$): $\rho_{crit} = 64$, $\lambda = 0.046$. S0-4 ($a = 2.4$): $\rho_{crit} = 64$, $\lambda = 0.046$; $\rho_{crit} = 512$, $\lambda = 0.037$. S0-4 ($a = 3.0$): $\rho_{crit} = 64$, $\lambda = 0.048$; $\rho_{crit} = 512$, $\lambda = 0.039$. S0' ($a = 3.0$): $\rho_{crit} = 64$, $\lambda = 0.058$. S0'' ($a = 6.0$): $\rho_{crit} = 64$, $\lambda = 0.042$. W0-1 ($a = 3.88$): $\rho_{crit} = 64$, $\lambda = 0.045$. W0-1 ($a = 9.54$): $\rho_{crit} = 64$, $\lambda = 0.052$. W0-1 ($a = 23.4$): $\rho_{crit} = 8$, $\lambda = 0.058$; $\rho_{crit} = 64$, $\lambda = 0.052$; $\rho_{crit} = 512$, $\lambda = 0.045$.

Log-linear regression across all 17 B&E data points gives $r = -0.77$, $p = 0.0003$. The ρ_{crit} column in Table 4 is explicitly stated in units of mean density $\bar{\rho}$ in the Table 4 notes: “The resulting groups are approximately bounded by surfaces of constant density ρ_{crit} given in units of the mean density.” No conversion is therefore required to place these points on the Figure 4 x-axis.

B.4 MDPL2 — X-Axis Derivation

Source: CosmoSim public database, MultiDark Project. Accessed March 2026.
<https://www.cosmosim.org>

Simulation: MDPL2 (MultiDark Planck 2), pure DM-only, TreePM gravitational solver with FFT long-range forces. Box side length 1000 Mpc/h, 3840^3 particles, Planck cosmology ($\Omega_m = 0.3089$). Halos identified using Rockstar at $z = 0$.

Query: SELECT AVG(spin_bullock) FROM mdpl2.rockstar WHERE upid = -1 AND spin_bullock > 0 AND mvir > 1e11 AND scale = 1.0. Result: N=30,776,052 distinct halos, $\lambda_{mean} = 0.0416$, $std = 0.026$.

The x-axis position at $z = 0$ is: $\rho_{crit} = 200 \times E^2(z = 0) / \Omega_m = 200 \times 1.0 / 0.3089 = 647 \times$ mean background density at $z = 0$. This is an independent code confirmation point at the same overdensity region as Benson (2017) at $712 \times$. Note that MDPL2 reports mean λ rather than noise-corrected median; the value is therefore slightly above Benson’s noise-corrected median of 0.0369, as expected. Both points use a TreePM FFT solver and share the $k \times k = 0$ structural property identified in Appendix A.

B.5 Benson (2017) — X-Axis Derivation

Source: Benson, A. J. (2017). Constraining the noise-free distribution of halo spin parameters. MNRAS, 471, 2871–2881.

Simulation: Millennium simulation, GADGET-2 TreePM solver, Millennium cosmology (WMAP1: $\Omega_m = 0.25$, $\Omega_\Lambda = 0.75$). Halos identified using Friends-of-Friends (FoF) with linking length $b = 0.2$ at $z \approx 0$.

Reported value: $\lambda = 0.0369$ (noise-corrected median)

By convention, the Friends-of-Friends (FoF) halo finder with linking length $b = 0.2$ is taken to approximate the virial overdensity predicted by spherical collapse in an Einstein–de Sitter universe, which is $178 \times \rho_{crit}$ at the redshift of identification. To express this threshold in units of the mean background density at $z = 0$ (the x-axis of Figure 4), we convert as follows:

Step 1. At $z = 0$, $E^2(z = 0) = 1.0$ by definition.

Step 2. The FoF $b = 0.2$ threshold corresponds conventionally to $\Delta \approx 178 \times \rho_{crit}(z = 0)$. To express this in units of the mean background density ρ_{mean} :

$$\frac{\rho_{crit}}{\rho_{mean}} = \frac{1}{\Omega_m} = \frac{1}{0.25} = 4$$

Thus:

$$178 \times \rho_{\text{crit}} = 178 \times 4 \times \rho_{\text{mean}} = 712 \times \rho_{\text{mean}}$$

Result: The Benson point is placed at $x = 712$ (units of mean background density) with $\lambda = 0.0369$.

Note: Benson (2017) does not explicitly state an overdensity threshold in units of mean background density. The x-axis position above is derived from the standard FoF $b = 0.2$ convention, which by long-standing practice is taken to approximate the virial overdensity predicted by spherical collapse in an Einstein–de Sitter universe, i.e., $178 \times \rho_{\text{crit}}$ at the redshift of identification.

In the Millennium cosmology ($\Omega_m = 0.25$, $\Omega_\Lambda = 0.75$), the true virial overdensity at $z = 0$ differs from this conventional value. Using the Bryan & Norman (1998) fitting formula for a flat universe:

$$\Delta_{\text{vir}}(z) = 18\pi^2 + 82x - 39x^2, \text{ where } x = \Omega_m(z) - 1$$

At $z = 0$, $\Omega_m(0) = 0.25$, so:

$$x = 0.25 - 1 = -0.75$$

Plugging in:

$$\begin{aligned} \Delta_{\text{vir}} &= 18\pi^2 + 82(-0.75) - 39(-0.75)^2 \\ 18\pi^2 &\approx 18 \times 9.8696 = 177.65 \\ 82 \times (-0.75) &= -61.5 \\ (-0.75)^2 &= 0.5625, 39 \times 0.5625 = 21.9375 \end{aligned}$$

Summing:

$$\Delta_{\text{vir}} \approx 177.65 - 61.5 - 21.9375 = 94.21$$

Thus, the true virial overdensity is approximately:

$$\Delta_{\text{vir}} \approx 94 \times \rho_{\text{crit}}(z = 0)$$

To express this in units of the mean background density at $z = 0$:

$$\frac{\rho_{\text{crit}}}{\rho_{\text{mean}}} = \frac{1}{\Omega_m} = \frac{1}{0.25} = 4$$

Therefore:

$$94 \times \rho_{\text{crit}} = 94 \times 4 \times \rho_{\text{mean}} = 376 \times \rho_{\text{mean}}$$

If this true value were used, the Benson point would shift from $712 \times \rho_{\text{mean}}$ (conventional) to $376 \times \rho_{\text{mean}}$ (physical). This leftward shift moves the point to a lower overdensity region where the B&E log-linear trend predicts a higher λ than at $712\times$. Since the observed value of 0.0369 is already below the trend at the conventional position, the physically corrected placement increases the negative residual, deepening rather than relieving the discrepancy.

The conventional mapping ($712 \times \rho_{\text{mean}}$) is retained in Figure 4 for consistency with the extensive literature that uses FoF $b = 0.2$ as a virial mass proxy without cosmology-specific correction. Importantly, using the true value would only *steepen the observed decline* and further strengthen the conclusion that $\mathbb{E}[\lambda] \rightarrow 0$.

B.6 Li et al. (2022) — X-Axis Derivation

Source: Li, J., Obreschkow, D., Lagos, C., Elahi, P., & Stevens, A. (2022). Spin transfer in collapsing haloes. MNRAS, 515(1), 437–450.

Simulation: SURFS, non-radiative run (L210N1024NR), GADGET-2. Halos identified at $z = 4$ using VELOCIRAPTOR with virial definition Δ_{200} relative to mean background density at $z = 4$. SURFS cosmology: $\Omega_m = 0.3121$, $\Omega_\Lambda = 0.6879$, $h = 0.6751$.

Reported value: $\lambda_{DM} = 0.0255$ (dark matter only component)

Li et al. use Δ_{200} relative to mean background density at $z=4$, not relative to critical density. Their virial radius is defined as the radius of a spherical region with mean interior density equal to 200 times the mean background density at $z = 4$. Since mean background density scales as $\rho_{mean}(z) = \rho_{mean}(z = 0) \times (1 + z)^3$, the conversion to x-axis units is cosmology-independent:

Step 1. The x-axis position expressed relative to mean background density at $z = 0$:

$$\begin{aligned}\rho_{crit}(x\text{-axis}) &= 200 \times \rho_{mean}(z=4) / \rho_{mean}(z=0) = 200 \times (1+z)^3 = 200 \times (1+4)^3 \\ &= 200 \times 125 = 25,000 \times \text{mean}\end{aligned}$$

Step 2. This result is cosmology-independent, meaning that it follows solely from the mean density scaling relation and the redshift of the snapshot.

Li et al. x-axis position = 25,000 × mean. Reported as $\approx 25,000 \times$ in the main paper.

Result: $\lambda_{DM} = 0.0255$ at $z = 4$ from Table 1 of Li et al. (2022), dark matter only component.

Note: In v2 of the main paper, the Li et al. x-axis was incorrectly reported as $\rho_{crit} \approx 1800 \times \text{mean}$. This was corrected in v3 using the derivation above.

B.7 Shin-Uchuu (Ishiyama et al. 2021) — X-Axis Derivation

Source: Ishiyama, T. et al. (2021). The Uchuu simulations: Data Release 1 and dark matter halo concentrations. MNRAS, 506(3), 4210–4231.

Simulation: Shin-Uchuu, pure DM-only. 262 billion particles. Halos at $z = 3.93$ using Δ_{200C} . $M_{200C} > 10^9 M_\odot/h$.

Reported value: $\lambda_{DM} = 0.0316$ (median), from 5,292,437 distinct halos.

Script: `lambda_conc.py`, available in the GitHub repository.

Step 1. Compute $E^2(z = 3.93)$:

$$E^2(z=3.93) = 0.3089 \times (1+3.93)^3 + 0.6911 = 0.3089 \times 119.095 + 0.6911 = 37.72$$

Step 2. X-axis position:

$$\rho_{crit}(x\text{-axis}) = 200 \times E^2(z=3.93) / \Omega_m = 200 \times 37.72 / 0.3089 = 24,412 \times \text{mean}$$

Result: Shin-Uchuu x-axis position = 24,412 × mean. Reported as $\approx 24,400 \times$ in the main paper.

B.8 Script Output: `lambda_conc.py` (Shin-Uchuu)

The `lambda_conc.py` script reads the Shin-Uchuu halo catalog at $z = 3.93$ and computes median λ as a function of overdensity threshold. The relevant output (`ShinUchuu_lambda_od.csv`) is:

<i>od_threshold</i>	<i>N_halos</i>	<i>median_λ</i>	<i>mean_λ</i>	<i>std_λ</i>	<i>z</i>	<i>sim</i>
1,800	5,292,437	0.0316	0.0358	0.0217	3.93	ShinUchuu_DM
5,000	5,292,437	0.0316	0.0358	0.0217	3.93	ShinUchuu_DM

The median $\lambda = 0.0316$ is stable across overdensity thresholds from 1,800 to 5,000 $\times$ mean, confirming the result is not sensitive to the exact threshold definition.

B.9 Summary of All Data Points

<i>Dataset</i>	<i>z</i>	<i>λ</i>	<i>x – axis</i> ($\times$ <i>mean</i>)	<i>Method</i>
Barnes & Efstathiou (1987)	~ 0	0.037–0.058	8–512	Direct from Table 4
Benson (2017)	~ 0	0.0369	712	FoF $b = 0.2 \rightarrow 178 / \Omega_m$ (Millennium cosmology)
MDPL2	0	0.0416 (mean)	647	$200 \times E^2(z = 0) / \Omega_m$, Planck, CosmoSim query
Li et al. (2022)	4	0.0255	25,000	$200 \times (1 + z)^3 - \Delta_{200}$ mean density definition
Shin-Uchuu (2021)	3.93	0.0316	24,412	$200 \times E^2(z = 3.93) / \Omega_m$ (Planck)

No free parameters enter the placement of any data point. The B&E x-axis values are taken directly from Table 4 in units of mean density $\bar{\rho}$. The Li et al. and Shin-Uchuu positions follow from their virial definitions. The Benson point is derived from FoF $b = 0.2$ convention. The log-linear trend $r = -0.88$ across all four datasets is a consequence of the data, not of any tuning of the axis placement.

B.10 References

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